
Typical: Direct Typed Decisions from Pretrained Language Models Without Answer Generation

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Abstract

Software that calls a language model to make a bounded decision – route this ticket, flag this transaction, score this alert – usually recovers the decision from generated answer symbols, constrained decoding, or candidate log-probabilities, even though what the caller needs is a distribution over a label set defined at request time. We ask whether a pretrained causal language model can expose that decision *directly*, without generating an answer, and what must remain inside its computation for the decision to stay question-conditioned. Our central result is negative and scoped. A decision state computed *before* the candidate set is known – the formulation that would make one state reusable across arbitrary candidate sets – transfers a listwise teacher’s candidate priors and calibration but not its question-conditioned closed-book capability. This holds across six tested factorizations and distillation objectives and at both backbone depths we tapped, and it is invisible to raw accuracy: it appears only under Δ_q , the drop in accuracy when the real question is replaced by a shuffled one with the candidate set held fixed. **Typical** instead renders the query and its runtime-defined candidates inside one causal suffix against a state whose key-value cache is computed once, and scores the terminal decision state against each candidate’s own *contextual* hidden state. That readout recovers the question-conditioned signal of a letter-generating interface (Δ_q^{sh} .106 vs. .109; teacher .102) with about a quarter of its order sensitivity, and carries typed output contracts – categorical Choice, ordinal Score, Bernoulli Noul – plus an abstention gate factored out of the candidate softmax. Standard bounded decisions generally improve with backbone scale (1.7B–14B, with an anomalous 8B point) and soft-target probability quality improves substantially at fixed capacity through objective and checkpoint-selection changes. The boundary is sharp and we report it as prominently: hard compositional decisions remain unresolved at every size we trained, and a same-size *frozen* backbone read with three in-context exemplars is not beaten there by any model we fine-tuned.

1 Introduction

Software increasingly asks a language model to make a decision, not to write a paragraph: route this ticket to one of forty queues, decide whether this transaction is fraudulent, score the severity of this alert, choose the next tool call. In these settings the caller ultimately needs a structured decision – often a probability distribution over a label set defined at request time – rather than free-form text. Existing approaches typically recover that decision through generated answer symbols or strings, constrained decoding, or candidate log-probabilities, which introduces decoding cost and a sensitivity to the answer representation that is well documented: option-order and option-symbol biases, and first-token probabilities that disagree with the text the model would actually generate.

This paper asks one question:

Can a pretrained causal language model be read directly as a decision engine, without generating an answer at all – and what has to remain inside its computation for that to work?

We break it into three research questions, which organize the rest of the paper.

- RQ1** Can a decision representation be computed *before* the candidate set is known and still preserve question-conditioned language-model capability? (Section 5.1)
- RQ2** If not, can candidate-aware contextual states expose the decision directly, without answer generation? (Sections 5.2 to 5.4)
- RQ3** How does such an interface behave under scaling, calibration training, and harder compositional decisions? (Sections 6 and 7)

RQ1 matters because a candidate-blind decision state is the formulation a systems designer wants: one state, computed once, reusable against any candidate set a runtime enumerates later, with scoring cost independent of the language model. We tested it and it does not work, in a specific and measurable way. Under six tested factorizations and distillation objectives, and at both backbone depths we tapped, a candidate-blind state absorbs a listwise teacher’s candidate-set priors and calibration but not its question-conditioned closed-book capability. That failure is invisible to accuracy: the students score well above chance. It appears only under a control that replaces the real question with a shuffled, unrelated one and checks whether the score moves. We make that control a first-class metric, Δ_q (Section 4.3), and report the negative as load-bearing rather than as a discarded pilot – it is the reason the architecture we adopt gives up candidate-independent scoring.

The answer to RQ2 is that the candidates must participate in the language model’s own computation, and that *how* they are represented in the readout matters at the interface level. Typical encodes a state once and retains its key-value cache; every query against that state is a short causal suffix – query text and candidate labels together – attending to the cached prefix and never recomputing it. A direct readout then scores the terminal decision state against each candidate’s own *contextual* hidden state. Scoring instead against a candidate-blind semantic embedding of the same option text loses 13 points of knowledge-probe accuracy; scoring against the contextual state matches a letter-generating interface’s question-conditioned signal with roughly a quarter of its order sensitivity. Recent mechanistic work shows a decoder already knows which multiple-choice option is correct before it emits the answer symbol [Wong et al., 2026]; we build on that finding rather than restate it, by making the candidate-aware latent representation the deployed output interface and measuring what is lost when the candidates are removed from the computation. Figure 1 sketches the resulting decision pass and the typed output contracts on it.

RQ3 gives the boundary. On standard, bounded decisions the recipe generally improves with backbone size across the tested Qwen3 checkpoints, and soft-target probability quality improves substantially at fixed capacity once checkpoints are selected on a held-out calibration metric rather than in-distribution loss. On a harder tier of compositional, multi-step decisions nothing we trained clears a same-size *frozen* backbone read with three in-context exemplars. We report that comparison with its own confound: state and option *rendering* alone moves a frozen backbone’s score by a margin comparable to the effect being discussed (Section 7.3).

Contributions. Three primary, one diagnostic.

1. **A candidate-blind negative result.** Under six tested factorizations and distillation objectives, and at both tapped depths, candidate-blind decision states transfer candidate priors and calibration but not question-conditioned closed-book capability (Section 3.4, Section 5.1). The shuffled-question control that exposes it, formalized as Δ_q , separates genuine question dependence from candidate-set priors and is itself a reusable diagnostic. We state this as a scoped empirical finding – under these factorizations, objectives, depths, and budget – not an impossibility theorem.
2. **A candidate-aware direct readout.** A cached-prefix decision pass scores a terminal decision representation against each candidate’s contextual hidden state, recovering the question-conditioned signal of a letter-generating interface while reducing order sensitivity and removing answer-token generation entirely (Section 3.2, Section 5.2). Typed output contracts (categorical, ordinal, Bernoulli) and an abstention gate factored out of the candidate softmax are components of that interface (Section 3.3, Sections 5.4 and 6.1).

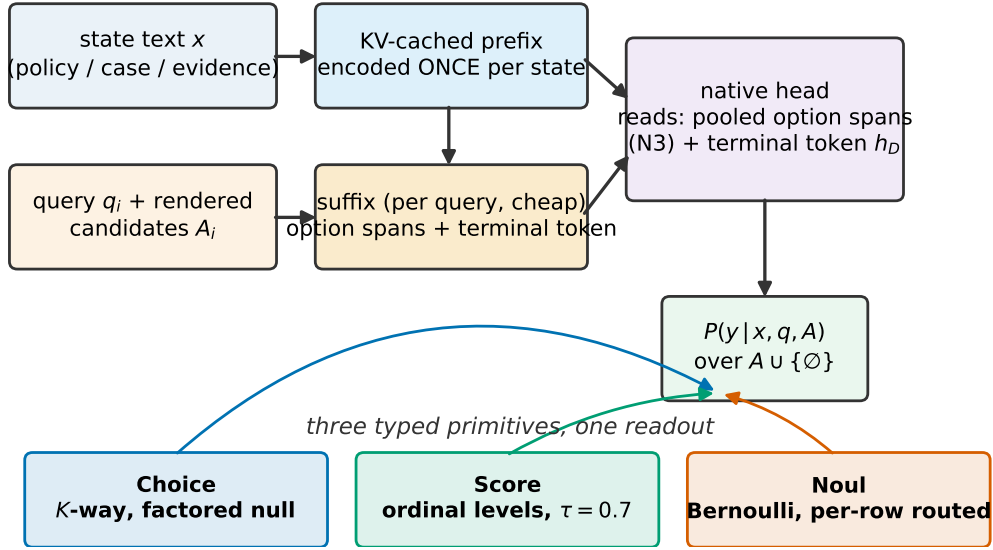


Figure 1: Schematic of the decision pass: KV-cached state prefix, per-query suffix, native head reading pooled option spans + terminal token, three typed primitives (Choice / Score / Noul) on one readout.

3. **A capability and scaling boundary.** Across matched scales and frozen controls, standard bounded decisions generally improve with scale and soft-target likelihood improves without additional capacity; hard compositional decisions remain unresolved and are not better after fine-tuning than under frozen few-shot inference (Section 6.3, Sections 7.1 and 7.2).
4. **Diagnosed failure modes** that aggregate accuracy hides: a last-layer tap that discards the state, a rendered abstain option that dilutes under softmax competition as the candidate set grows, and an evidence-removing truncation defect in long-context training data whose cost we isolate from a render-compatibility confound with a 2×2 ablation replicated at two scales (Section 5.3, Section 5.4, Section 7.4). We treat these as secondary contributions supporting the three claims above.

2 Related work

Runtime-defined label spaces. Deciding over a candidate set supplied at inference time is an old problem. GILE embeds inputs and label descriptions in a shared space so unseen labels are scorable [Pappas and Henderson, 2019]; entailment reformulation turns arbitrary label names into hypotheses [Yin et al., 2019]; TARS conditions a sentence encoder on the label text itself [Halder et al., 2020]; GLiNER and GLiNER2 encode text and a runtime schema jointly in one bidirectional pass [Zaratiana et al., 2024, 2025], the latter also appearing on the JevBench leaderboard we report against. The cost axis is the classical cross- versus dual-encoder trade-off: a cross-encoder reranker attends candidate to input and pays one forward pass per candidate [Nogueira and Cho, 2019], a dual encoder is candidate-independent but interacts only at a dot product [Reimers and Gurevych, 2019], and late-interaction and decomposition methods sit between – Poly-encoders [Humeau et al., 2019], ColBERT [Khattab and Zaharia, 2020], DeFormer [Cao et al., 2020a], PreTTR [MacAvaney et al., 2020], LUMEN [de Jong et al., 2023] – with compiling a cross-encoder into a cheaper student as standard practice [Lu et al., 2022]. Decoder LLMs with a classification head close the loop [Li et al., 2023]. Four distinctions are worth making explicit. Against *fixed classifiers and sequence classification*: our candidates are runtime-defined text, not fixed class indices. Against *GILE, TARS, and dual encoders*: our central negative result is precisely that a candidate-independent representation is insufficient for the targeted question-conditioned closed-book regime (Section 5.1). Against *cross-*

encoders and rerankers: Typical reuses one state key-value prefix across many decisions and emits a typed distribution with an abstention event, rather than independently reranking state–candidate pairs. Against *poly-encoders*, *ColBERT*, and *late interaction*: our candidates are rendered inside the causal computation and acquire contextual slot and set information that a late-interaction embedding of the candidate in isolation does not carry. Our own energy-scoring shortlisting front end (Section 6.5) is a member of this family and is retained for exactly the sub-task – bounded- K pre-filtering – that does not require question-conditioned capability.

Reading the decision out of hidden states instead of generating it. This is our closest prior art. Linear probes and their controls establish that hidden states carry decodable structure [Alain and Bengio, 2016, Hewitt and Liang, 2019, Belinkov, 2022]; layerwise analysis shows different depths carry different abstractions [Tenney et al., 2019, Belrose et al., 2023]; states anticipate tokens not yet emitted [Pal et al., 2023, Wu et al., 2024]; and MCQ-specific circuit analysis localizes the symbol-binding step [Lieberum et al., 2023]. Most directly, Wong et al. [2026] show a two-stage computation in ordinary multiple-choice prompting: the winning option is decodable in content space at the option boundary *before* the answer symbol is bound, a finding traced across languages by Bhatt et al. [2026]. The letter interface itself is documented as unreliable: selection bias and option-order sensitivity [Zheng et al., 2023, Pezeshkpour and Hruschka, 2024], symbol-identity effects and binding fixes [Yang et al., 2025b, Xue et al., 2024], first-token probabilities that disagree with the generated answer [Wang et al., 2024a], and the choices-only control quantifying how much multiple-choice accuracy needs no question at all [Balepur et al., 2024] – the control we adopt as our primary diagnostic (Section 4.4). Prior work observes the latent decision and diagnoses the symbol interface. We do not claim novelty at the level of “the answer is latent before generation” – that is Wong et al. [2026]. What we contribute is to make the candidate-aware latent representation the *deployed output interface* and to measure what is lost when the candidates are removed from the computation, including that the last layer is the wrong tap (Gromov et al., 2024, Men et al., 2025, Schwartz et al., 2020; Section 5.3) and that the readout must be over contextual option spans, not slot-agnostic embeddings (Section 5.2).

Calibration, distillation, and what survives a change of interface. Temperature scaling largely fixes in-distribution expected calibration error [Guo et al., 2017], though binned ECE is itself a biased estimator [Pakdaman Naeini et al., 2015, Kumar et al., 2019], motivating proper scoring rules as the honest target [Gneiting and Raftery, 2007, BRIER, 1950], with the ranked probability score as their ordinal-aware member [Epstein, 1969, MURPHY, 1970]. Training-time levers include label smoothing [Szegedy et al., 2016, Müller et al., 2019] and soft-target distillation, which transfers more than argmax [Hinton et al., 2015, Kim and Rush, 2016] and improves probability quality even without a stronger teacher under the label-smoothing view of KD [Yuan et al., 2020, Zhang and Sabuncu, 2020]; fine-tuning specifically can distort calibration through its interaction with prior knowledge [Wang et al., 2025, Desai and Durrett, 2020]. Our calibration claims are training-time rather than post-hoc: soft and ordinal-smoothed targets (Section 3.3), a Brier term in the objective, and checkpoint selection on a held-out calibration metric (Section 3.6) rather than a single global temperature fit. Our frozen-teacher distillation term (Section 7.2) is a distillation target in this sense but a narrow one – the teacher is the same backbone read with a fixed few-shot rendering, so the relevant question is componential: which parts of a teacher’s behavior survive a change of interface. Our answer (priors and calibration, not question-conditioned knowledge) is the negative half of this literature, consistent with context distillation compiling a prompt into weights [Askell et al., 2021, Snell et al., 2022].

Prefix sharing and the cost of removing the decode loop. Sharing a long prefix across many suffixes is established systems work [Juravsky et al., 2024, Zheng et al., 2024, Kwon et al., 2023, Gim et al., 2023, Pope et al., 2022, Dao et al., 2022, Cheng et al., 2023]; generation-side accelerators are an informative contrast, since speculative decoding and Medusa exist because each sequential decode step is overhead-bound rather than compute-bound [Leviathan et al., 2022, Chen et al., 2023, Cai et al., 2024]. We contribute no new kernel; we report the measured, single-stream-harness consequence of deleting the decode loop entirely (Section 6.5).

Is generation necessary? Expressivity and latent reasoning. Chain-of-thought buys serial computation that a constant-depth transformer provably lacks in one forward pass [Wei et al., 2022, Li et al., 2024, Merrill and Sabharwal, 2023b,a, Feng et al., 2023], and helps empirically mainly on

math and symbolic tasks [Sprague et al., 2024]; filler and pause tokens show part of the gain is extra computation rather than verbalized reasoning [Goyal et al., 2023, Pfau et al., 2024], and implicit or latent chain-of-thought internalizes the steps [Deng et al., 2023, 2024, Hao et al., 2024]. This literature delimits our scope (Section 8): a single non-generative forward pass is the right instrument for bounded, evidence- or prior-driven decisions, and is predicted to fail on serial composition – exactly where our hard tier (temporal, probability, trade-off families) remains near chance.

Ordinal regression, abstention, and long-context failure modes. Our Score primitive is the minimal member of the ordinal-regression family [McCullagh, 1980, Frank and Hall, 2001, Niu et al., 2016, Cao et al., 2020b, Shi et al., 2023b, Díaz and Marathe, 2019]: a soft, ordinal-smoothed target that changes no decision and fixes probability quality, where a cumulative-link head did not win at matched budget (Section 6.1). Our abstain option is architectural rather than a classical reject rule [Chow, 1970, Geifman and El-Yaniv, 2017, 2019, Hendrycks and Gimpel, 2016, Liu et al., 2020], with out-of-scope intent detection as the applied benchmark [Larson et al., 2019, Casanueva et al., 2020]. On long context, retrieval position and length are known to change and degrade answers [Liu et al., 2024, Levy et al., 2024, Shi et al., 2023a, Hsieh et al., 2024]; our truncation finding (Section 7.4) is the training-side analogue, an instance of data-pipeline failures propagating silently through ML systems [Sambasivan et al., 2021].

Fine-tuning versus frozen backbones with in-context examples. Fine-tuning can distort pre-trained features and underperform out of distribution [Kumar et al., 2022], fair comparisons of few-shot fine-tuning against in-context learning are close [Mosbach et al., 2023], LoRA both learns and forgets less than full fine-tuning [Biderman et al., 2024, Hu et al., 2021] yet continual fine-tuning still forgets [Luo et al., 2025], and compositional-generalization benchmarks show where learned grammars stop composing out of distribution [Lake and Baroni, 2017, Keysers et al., 2019, Kim and Linzen, 2020, Hupkes et al., 2020]. Two of our findings live here: rubric-conditioned training transfers inside its own rule grammar and not to unseen composition (Section 7.1), and our trained 14B leads its frozen self on the standard tier while a same-size frozen backbone with three exemplars still leads on the hard tier (Section 7.2).

Typed decision benchmarks and commercial systems. JevBench, one of the evaluation suites we use, targets a class of commercial “System-One” decision models – most visibly TypeSafe’s Jev, documented only by vendor pages and a third-party black-box probe, closed-weight and closed-data. We therefore say that Typical targets a similar interface and task class, not that it reproduces Jev’s architecture, which we cannot inspect. The leaderboard carries several open replications that publish weights and scores but, apart from GLiNER2 [Zaratiana et al., 2025] and the ModernBERT backbone used by Laya [Warner et al., 2025], no method papers.

What we claim as new. Not “reading decisions from hidden states”, which has ample prior art above. The defensible contribution is narrower and sharper: we study the design space for turning a pretrained causal LM into a reusable typed decision primitive under a cached-state constraint, and identify a specific empirical boundary in it – candidate-blind reusable decision states fail to preserve question-conditioned capability, while candidate-contextual readout succeeds – together with the diagnostic (Δ_q) that makes the boundary visible at all. A secondary methodological contribution is that rendering is a first-class experimental factor for this task class: a frozen Qwen3.5-9B checkpoint moves 12.5 points of standard-tier accuracy on rendering choice alone (Section 7.3). We draw the cautious conclusion from that, not the convenient one: it means cross-entry benchmark comparisons in this area conflate model capability with interface compatibility – *ours included* – and that rendering should be held fixed when architectures are compared, not that any particular published result is wrong.

3 Method

Typical turns a pretrained causal language model into a decision engine: it consumes a runtime state, a query, and a runtime-defined candidate set, and returns a typed probability distribution over the candidates without generating any tokens. This section defines the task (Section 3.1), the decision pass (Section 3.2), typed outputs and abstention (Section 3.3), the candidate-blind alternative we tested and rejected (Section 3.4), and training (Section 3.5, Section 3.6). Throughout, we use

descriptive names for the three readouts compared in Section 5: the *letter-logit readout* (**N1** in tables), the *candidate-blind semantic readout* (**N2**), and the *contextual-candidate readout* (**N3**), which is what Typical deploys.

3.1 Task definition

Definition 3.1 (Typed decision). A decision is a tuple $D = (x, q, A, t)$: a state x (the document, transcript, or context the decision is about), a query q (the instruction or rubric), a candidate set $A = \{a_1, \dots, a_K\}$ of label *strings* supplied at request time rather than fixed at training time, and a primitive type t . The model returns

$$P(a_j \mid x, q, A) \text{ for } j = 1..K, \quad \text{and} \quad P(\emptyset) = P(\emptyset \mid x, q, A),$$

with no answer token emitted. The type t is a contract on the output distribution, not merely a different set of label strings: **Choice** is categorical over A ; **Score** is categorical over an *ordered* A with an ordinal-smoothed target, so probability mass adjacent to the gold level is not penalized as if it were an unrelated label; **Noul** is Bernoulli over a single proposition, order-invariant by construction; and $\emptyset$ is a separate abstention event, not a candidate.

3.2 The decision pass

Cached state, causal suffix. x is encoded once as an ordinary causal prefix and its key-value cache is retained; every query against it is a short causal suffix – query text followed by candidate labels – run through the backbone attending to the cached prefix, never recomputing it. This is exactly a forward pass on $\text{concat}(x, q, A)$ with the x half read from cache; a bit-exact equivalence test (state-KV-cache decode vs. full concatenation, tolerance 10^{-3} over chunk sizes 1, 2, 3, 5, 64) confirms no query recomputes the state and no two queries attend to each other. M decisions against one state therefore cost $O(x) + M \cdot O(q + A)$ rather than $M \cdot O(x + q + A)$: the state is amortized, but the per-query term still scales with the candidate tokens rendered in the suffix, not with q alone.

The contextual-candidate readout. The readout Typical deploys renders each candidate as lettered text inside the suffix (e.g. “A. *cancel_subscription* B. *escalate_to_human*”) and takes two things from one forward pass: the terminal hidden state h_D , and each candidate’s own contextual hidden state c_j^{ctx} , span-pooled over its rendered tokens. Because the candidates are tokens inside the same causal computation as the state and the query, c_j^{ctx} is not a semantic embedding of the label string: it carries slot identity and candidate-set context that a candidate embedded in isolation (a dual encoder or an energy-scoring front end) cannot represent. The score for candidate j is a bilinear function of h_D and c_j^{ctx} . Section 5.2 compares this against the letter-logit readout, which scores h_D against the static embedding of each rendered option letter, and the candidate-blind semantic readout, which scores h_D against an independently computed semantic embedding of the option text. We do not claim slot identity is the sole causal mechanism; we claim the contextual state contains slot identity and set context, and that ablating it to an independent semantic embedding is measurably insufficient.

3.3 Typed outputs and abstention

The null is factored out of the candidate softmax. A rendered “none of the above” option competing for softmax mass dilutes as K grows, independent of whether the true label is absent (Section 5.4). We instead compute $P(\emptyset)$ from a gate over permutation-invariant statistics of the candidate scores $s_1, \dots, s_K$ (max, top-2 margin, mean, log-sum-exp, each over valid candidates only) concatenated with h_D and the candidate-set mean/variance: $r = \sigma(g(z))$, giving

$$P(a_j \mid x, q, A) = (1 - r) \cdot \frac{\exp(s_j/T)}{\sum_{k=1}^K \exp(s_k/T)}, \quad P(\emptyset) = r, \quad (1)$$

implemented as logits $\ell_j = \log p_j + \log(1 - r)$, $\ell_\emptyset = \log r$. Three properties follow, and they are distinct claims that should not be collapsed into one. (i) Because r never enters the candidate softmax, the *pairwise odds among non-null candidates*, p_j/p_k , are exactly independent of the rest of the set under this factorization – an exact pairwise non-null odds property, *not* a statement that the model as a whole satisfies independence of irrelevant alternatives. (ii) The null gate r is computed from candidate-set statistics, so candidate-vs-null probabilities *remain set-dependent*, by design: whether to abstain should depend on what was offered. (iii) T scales only the candidate softmax; r ,

computed from untempered statistics, is T -invariant. We call this the **factored null**, distinct both from a candidate-aware null competing inside the same softmax (an earlier, energy-scoring version of the architecture) and from a literal rendered “none of the above” option. Null calibration *across* K is a separate, measured property (Section 5.4), not a consequence of the factorization.

Typed heads. Choice, Score, and Noul share one backbone, tap, and readout, differing only in target distribution and terminal head, routed per row rather than by a global flag: Score uses an ordinal-smoothed target over the rendered levels, Noul uses a dedicated Bernoulli head over the query alone (no candidate text) that is exactly order-invariant by construction, and the factored-null gate of Equation (1) applies to both. Appendix A.5 gives the exact parameterization, the equations, and an operational lesson from a routing bug that collapsed an early release candidate to chance.

3.4 The candidate-blind alternative

The design a systems engineer would prefer computes a decision state $Z(x, q)$ *before* the candidate set is known and scores candidates against Z afterward – a dual-encoder, energy-scoring-style factorization. It would let a runtime enumerate arbitrary candidates at a cost independent of the language model, and reuse one state across candidate sets rather than one prefix across suffixes. RQ1 asks whether such a state preserves question-conditioned capability. It is the hypothesis Section 5.1 tests and rejects.

The design cannot be refuted on accuracy alone, which is the methodological point: a candidate-blind state can absorb candidate-set priors from the training corpus and score well above chance while carrying no question-dependent signal at all. We therefore distilled a listwise teacher (a fine-tuned decoder reading options in context) into Z under several student architectures – a single vector; a multi-probe vector with token-level max-similarity scoring; a set-conditioned variant with $O(K)$ cross-attention over candidate tokens – and objectives – ordinary single-set distillation; an explicit same- Z multi-set objective supervising the *difference* in log-odds a candidate-set edit induces in the teacher – and measured each with Δ_q (Section 4.3) rather than accuracy.

3.5 Training

The recipe applies LoRA [Hu et al., 2021] (rank 16) to the top eight layers of a Qwen3 [Yang et al., 2025a] backbone truncated at roughly 71% of its depth (tap layer 20/28 at 1.7B, 26/36 at 4B/8B, 28/40 at 14B); layers below the tap are frozen. This truncation is not a compute optimization: reading the terminal state of the *last* layer degrades evidence-grounded decisions substantially (Section 5.3), since the final layer of a causal LM is specialized for next-token prediction rather than for accumulating the state’s evidence.

Batches are drawn from a **family-balanced sampler** over four families rather than a pool sampled by raw row count, so no single large corpus silently dominates: E (evidence and classification), K (closed-book *knowledge*), W (rubric-conditioned *workflow* decisions), U (soft-target *uncertainty*). The released recipe samples $E:0.40, K:0.15, W:0.35, U:0.10$ (Section 4.1 gives provenance); a sweep (Section 6.2) shows $E \geq 0.45$ is needed once W/U are added, with a wide flat region otherwise. The scientifically interesting result here is not the exact ratio but that the *distribution of decision types* materially controls capability, sometimes as much as architecture does (Section 8).

Null augmentation. 10–20% of W rows are duplicated with their gold candidate removed, target $\emptyset$, because the natural null rate differs by two orders of magnitude between corpora (under 1% of W rows carry one vs. over 20% of E) and a single global calibration offset cannot serve both regimes at once (Section 6.4): the offset that fixes false abstention on an evidence label space pushes a workflow-trained model to abstain on almost everything, and vice versa. Null augmentation fixes this at the data level.

Long-state handling. Long-policy workflow rows train at a longer window (up to 1,024 tokens for the release recipe, 3,072 for the 14B long-state candidate). An earlier version of the generator rendered supporting facts *after* the request text; because training right-truncates at the window, a 1,024-token window silently dropped the facts on 98.8% of long-policy rows. The fix re-renders **facts-first** and adds `-drop_truncated` to discard, rather than silently cut, any still-over-length row. Section 7.4 reports the matched ablation that isolates this from the render-compatibility effect it was confounded with, and measures what it cost.

Table 1: Training corpora. “Held out” counts whole families, domains, styles, or parameter ranges excluded from training entirely (not a random validation split); Table 2 maps these onto generalization axes.

Corpus	Train rows	Held out	Role
data_v5	633,213	label wordings	Evidence (NLI/BoolQ) + wide-vocabulary classification
data_kb	147,447	TruthfulQA-MC1, GPQA	Closed-book multiple-choice knowledge
data_wf	72,006	3 of 15 families	Rubric-conditioned workflow (Noul/Choice/Score)
data_wf_hf	86,600	3 external test/demo splits	External workflow sets (forms, rules, distillation)
data_wf_long	23,318	–	Long-state (multi-paragraph) workflow rows
data_wh	60,605	1 domain/level, 3 domains, 2 styles, level 7	DecisionMix v2 hard curriculum
data_u	31,029	4 synthetic parameter ranges	Soft-target uncertainty corpus

3.6 Checkpoint selection

Selecting checkpoints on in-distribution training loss favors the majority hard-label data at the expense of probability quality on the minority of soft and ordinal rows. We instead select on held-out uncertainty-and-curriculum negative log-likelihood (NLL), optionally with a Brier-score term in the training objective itself rather than only post-hoc temperature fitting. This change to the selection criterion alone – not model capacity or training data – produces the largest single swing in probability quality in this project (Section 6.4).

4 Experimental setup

4.1 Corpora

Training draws from seven corpora, summarized in Table 1: data_v5 (evidence – SNLI [Bowman et al., 2015], MNLI [Williams et al., 2018], ANLI [Nie et al., 2020a], BoolQ [Clark et al., 2019] – plus wide-vocabulary classification including CLINC-150 and Banking77/HWU64 [Larson et al., 2019, Casanueva et al., 2020]); data_kb (closed-book multiple-choice knowledge distilled from nine standard sources, TruthfulQA-MC1 [Lin et al., 2022] held out, leakage audited against MMLU-Pro [Wang et al., 2024b, Hendrycks et al., 2020]); data_wf and data_wf_hf (our rubric-conditioned workflow generator plus three externally sourced typed-decision sets); data_wf_long (a longer-state, facts-first variant of the workflow generator, Section 3.5); data_wh (DecisionMix v2, a second independent rubric-conditioned generator with compositional depth levels 1–7); and data_u (a soft-target uncertainty corpus from UNLI [Chen et al., 2020], AmbiEnt, and synthetic closed-form generators, with ChaosNLI-ambiguity [Nie et al., 2020b] evaluation-only). Full per-family breakdowns, licenses, and held-out split sizes are in Appendix A.8. None of the seven corpora contains any of the 231 public JevBench ids.

DecisionMix v2 deserves one note on its design, because it carries the paper’s strongest causal pressure on rubric conditioning: it emits *counterfactual rubric groups* – same state, same candidate set, different rubric, different gold – so a model cannot solve the task from the state or from candidate priors alone. The matching limitation belongs in the same breath: DecisionMix teaches transfer inside its own generated rule grammar. It does not establish general compositional rule execution, and its level-7 composition remains near chance for every model we trained (Section 7.1).

4.2 What “held out” means: a generalization taxonomy

“Held out” is used in this literature for at least seven materially different things, and conflating them is the fastest way to overread a runtime-defined-label result as arbitrary-domain zero-shot classification.

Table 2: Generalization taxonomy. Each row is a distinct axis with its own held-out split; none is a random validation split. The right-hand column is the scope a positive result on that axis licenses.

Axis	Evaluated on	A positive result licenses
Unseen examples	Held-out rows of every trained corpus	In-distribution competence only
Unseen label word- ing	data_v5 label paraphrases; CLINC-150 re-worded labels	Robustness to label surface form, same label space
Unseen label space	Banking77-77, HWU64, tree-choice (K to 320), TruthfulQA-MC1	Scoring labels never trained on, within the trained domains
Unseen workflow family	3 of 15 data_wf families	Transfer across task families from one generator
Unseen rubric style	2 withheld data_wh writing styles	Rubric conditioning is not style memorization
Unseen grammar / composition	1 withheld domain $\times$ level grammar; DecisionMix level 7	Compositional rule execution — <i>not</i> obtained (Section 7.1)
External dataset	PagerDuty, jevlogs, Mind2Web, typed-decisions, JevBench	Transfer outside our generators entirely

Table 2 separates the axes we actually measure and states what a positive result on each does and does not license. Every claim in Sections 5 to 7 is tagged to one of these rows.

External, evaluation-only sets. Six sets are never trained on: **PagerDuty** (alert-triage yes/no, floor .792), **jevlogs** (log-triage yes/no, floor .697), **Mind2Web** (web-agent action choice, floor .427), **tree-choice** (taxonomy-routed choice up to $K=320$), **typed-decisions** (400 cases spanning all three primitive types with full instructions, criteria, and probabilistic gold), and **TruthfulQA-MC1**. We report accuracy against each set’s own majority-class floor because several are class-imbalanced enough that a constant prediction clears a nontrivial accuracy; a model below its floor is worse than guessing the majority class.

We additionally report **rubric-flip accuracy** – for a state and candidate set rendered under two rubrics with different gold answers, the fraction of pairs where the model gets *both* answers correct, only possible if the rubric conditions the answer – and $\Delta_r = \text{Acc}(x, r, A) - \text{Acc}(x, r_{\text{shuf}}, A)$, comparing accuracy under the correct rubric against a swapped one, state and candidates fixed.

4.3 Δ_q : separating question dependence from candidate priors

Definition 4.1 (Question-dependence gap). For a model f and an evaluation set of decisions,

$$\Delta_q = \text{Acc}(f(x, q, A)) - \text{Acc}(f(x, \tilde{q}, A)),$$

where $\tilde{q}$ replaces the real question while the candidate set A and the state are held fixed. We report two variants: a *choices-only* variant ($\tilde{q}$ is an empty placeholder) and the stricter *shuffled-question* variant Δ_q^{sh} ($\tilde{q}$ is a real but unrelated question from the same corpus), which is primary throughout, after the choices-only variant was found to leak residual candidate priors through the degenerate rendering.

The interpretation is the reason this metric exists:

- **High raw accuracy with $\Delta_q \approx 0$** means the model is exploiting candidate-set priors or answer-set artifacts. It is not answering the question. Accuracy alone cannot see this.
- $\Delta_q > 0$ means a question-dependent signal is present, and its magnitude is interpretable relative to a reference model measured on the same items.

The reference throughout is the listwise teacher’s own $\Delta_q^{\text{sh}} = .102$ on 1,200 MMLU-Pro items: the teacher’s question-dependent component is itself small, so the measurement has roughly nine points of dynamic range, not twenty-one. We treat differences under $\pm .02$ as indistinguishable from zero; that band is approximately 1.4 standard errors of an *unpaired* difference of proportions at $n=1,200$ and $p \approx .15$, and a paired comparison on identical items would be tighter still, so it is a conservative band. This metric is adopted from the choices-only control of Balepur et al. [2024], strengthened to a shuffled real question.

4.4 Metrics

We report **accuracy** (argmax against gold), **Brier score**, **expected calibration error (ECE)**, and **negative log-likelihood (NLL)**, against hard or soft gold as the evaluation set provides. Where a label space is only partially covered by the candidate set at test time, we separately report *among-K* accuracy (argmax restricted to the true candidate set, ignoring $P(\emptyset)$) so null and discrimination failures are not conflated. Rubric-conditioned score and typed external sets carry soft, non-one-hot gold, so NLL and Brier score are primary there and accuracy is a secondary, coarser signal. Calibration is reported as a *measured axis per task and type*, never as a blanket property of the model.

4.5 JevBench protocol, disclosures, and confidence intervals

JevBench (fstandhartinger/jevbench v1.2.1) is a third-party benchmark for typed decision models, with items typed `noul` (yes/no), `choice` ($K = 2-6$), and `score` (ordinal), each supplying a state, an instruction/rubric, and an exact label set, split into standard, easy, and hard tiers. It is one evaluation suite among several here, not the foundation of any claim.

Public subset, not a ranked entry. The harness’s public release exposes 231 of its items (72 standard, 48 easy, 111 hard); a further 146 judge-scored items are not public, and the harness’s own leaderboard ranks only submissions covering at least 95% of items including that private set. Every number we report is computed on the 231 public ids only and is *not* a ranked leaderboard entry. We carry this disclosure in the caption of every JevBench table and figure rather than in prose alone. Contextual comparison against other leaderboard entries is deferred to Appendix A.11, because protocols and rendering differ between entries, several entries have no method paper, the private judge items are unavailable, and our own rendering control (Section 7.3) moves scores by more than the spread being compared. The scientifically useful JevBench result in this paper is the within-model matched comparison, not a ranking.

Confidence intervals and the paraphrase cluster. The standard tier contains 72 items but only 36 *independent states*: each state appears as two paraphrases sharing a group field. Resampling items would treat paraphrase pairs as independent and report intervals that are too narrow, so we **cluster-bootstrap over group** (20,000 resamples); the hard tier has one item per group and reduces to an ordinary item bootstrap. The resulting 95% intervals are wide and should discipline every reading of this benchmark: roughly ± 6 to ± 13 points at the standard tier ($n_{\text{eff}}=36$) and ± 9 points at the hard tier ($n=111$). Where two of our own checkpoints are compared on the same items we additionally report a *paired* cluster bootstrap on the per-item difference, which is the appropriate test and is materially more powerful than comparing two marginal intervals. We do not treat any unpaired difference smaller than these intervals as a ranking.

Probabilities are conditioned on non- $\emptyset$. The harness’s scoring expects a distribution over the labeled candidates only; we report $P(a_j \mid x, q, A, \neg\emptyset)$, i.e. $P(\emptyset)$ is computed and then dropped and the remaining mass renormalized.

Other disclosures, held constant across every model we report: option order follows the harness’s own label order (a reversed-order control on one checkpoint swung standard accuracy by ± 4 points); latency is measured in-process on one H100, one decision at a time, with model loading excluded and cold candidate-label embedding included where applicable; and no hosted-API cost applies to any of our numbers. Every run’s harness commit, checkpoint hash, and dependency lock file are recorded in the run manifest.

4.6 Hardware and latency measurement

Training runs on rented NVIDIA H100 GPUs (SXM and NVL variants), bf16, AdamW, effective batch size 64, 8,000–12,000 steps depending on the run. Latency is measured in-process, single stream, one decision at a time unless a batch size is stated, with the state’s key-value cache warm and model loading excluded. Numbers compared across backbone sizes in one table are measured on one pod with one PyTorch/CUDA build; numbers from different pods or builds are marked as such and not compared directly. We keep two latency stories separate throughout: a controlled research-harness measurement (Section 6.5) and an optimized serving path with key-value reuse (Appendix A.3).

Table 3: Claim $\rightarrow$ evidence $\rightarrow$ scope. Section pointers are to the primary evidence, not to every mention.

Claim	Evidence	Scope / limitation
Candidate-blind states lose question dependence (RQ1)	6 factorizations/objectives, 2 depths, Δ_q^{sh} shuffled-question control (Section 5.1)	Not an impossibility theorem; one backbone family, 1.7B, 12k steps
Contextual candidates recover it (RQ2)	Contextual-candidate vs. letter-logit vs. candidate-blind semantic readouts, seed pair (Section 5.2)	Equivalence to the letter interface is at the seed-noise floor; still order-sensitive
Factored null fixes the rendered-abstain pathology	K -sweep, false-abstain rates (Section 5.4)	Null gate remains set-dependent by design; partly a property of our null rate in training
Scale helps standard decisions	Matched 1.7B–14B ladder with frozen controls (Section 6.3)	Anomalous 8B point; single seed per rung; a ladder, not a scaling law
Probability quality is trainable at fixed capacity	Checkpoint-selection and objective isolation (Section 6.4)	Type- and family-dependent; hard-tier calibration stays poor
Hard composition is unresolved	Frozen-backbone paired comparison, per-family breakdown (Section 7)	Rendering not controlled in the frozen comparison; data coverage not excluded
Truncating the evidence out of training data costs capability	2×2 train-render $\times$ eval-render ablation, 605 held-out states, replicated at 1.7B and 14B (Section 7.4)	Single-variable only at 1.7B; evaluation set shares a generator with training

4.7 Roadmap: claims, evidence, and scope

Table 3 is the map from the three claims of Section 1 to the evidence that supports each and the scope limitation attached to it. Section 5 answers RQ1 and RQ2, Section 6 answers the first half of RQ3, and Section 7 answers the second half. The chronological development log, which is how these experiments were actually run, is in Appendix A.10.

5 What must be in a direct decision representation?

This section answers RQ1 and RQ2. Section 5.1 shows that a candidate-blind decision state does not preserve question-conditioned capability; Section 5.2 shows that a contextual-candidate readout does; Section 5.3 and Section 5.4 isolate the two design choices that readout needs to work. Unless noted otherwise, JevBench numbers follow the disclosures of Section 4.5 (public 231-id subset, unranked, probabilities conditioned on non- $\emptyset$).

5.1 Candidate-blind decision states do not preserve question-conditioned capability

Figure 2 is the paper’s central negative result and the reason for everything that follows. The x -axis is raw accuracy on a 1,200-item MMLU-Pro among- K knowledge probe; the y -axis is Δ_q^{sh} (Definition 4.1). Every candidate-blind variant we trained lands in the $\pm.02$ band around zero, and four of the six are *below* zero: the real question buys them nothing over a shuffled, unrelated one with the candidate set held fixed. Their above-chance accuracy of .12–.17 is candidate-set priors baked into the corpus.

Table 4 gives the variants. The negative survives every lever we had. *Head capacity* is not the bottleneck: a single decision vector retains as much as eight probes. *Set conditioning* is not the bottleneck: adding $O(K)$ cross-attention over candidate tokens changes nothing on this axis (it does improve null quality, Section 5.4). *Supervision* is not the bottleneck: an explicit same- Z multi-set objective that supervises the *difference* in log-odds a candidate-set edit induces in the teacher reproduces the teacher’s Δ no better than its own steps-matched single-set control, and its Δ_q^{sh} is exactly zero. And *tap depth* is not the bottleneck: re-running the strongest student at full backbone depth, after depth turned out to be decisive for the candidate-*aware* readout (Section 5.3), leaves Δ_q^{sh} at $-.008$ while among- K accuracy *falls* to .147 and every evidence task regresses in the pattern a last-layer tap predicts (SNLI/MNLI/ANLI/BoolQ .881/.807/.462/.712 at full depth vs.

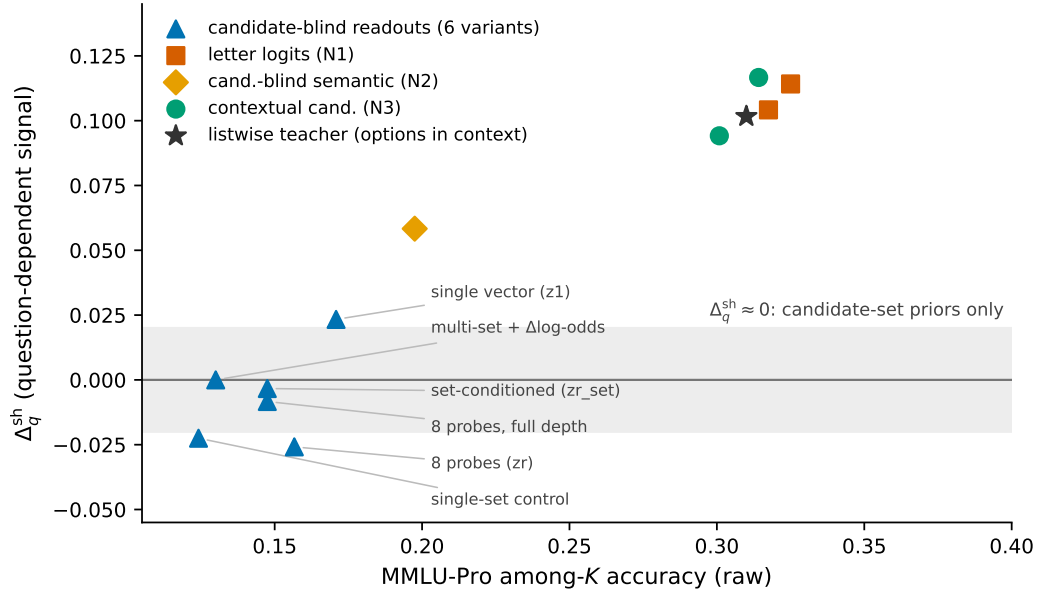


Figure 2: The candidate-blind negative. Raw MMLU-Pro among- K accuracy against Δ_q^{sh} , the drop in accuracy when the real question is replaced by a shuffled one with the candidate set held fixed ($n=1,200$ items; the shaded band is the ± 0.02 indistinguishable-from-zero region of Section 4.3). All six candidate-blind variants sit in that band regardless of head capacity, set conditioning, objective, or tap depth. The letter-logit readout, the contextual-candidate readout, and the listwise teacher all carry a question-dependent signal of .09–.12. Raw accuracy alone cannot tell these two groups apart.

Table 4: Candidate-blind students: six factorizations and objectives, two tap depths. “Shuffled” replaces the real question with an unrelated one, candidate set fixed. Δ_q^{sh} is the difference. $n=1,200$ MMLU-Pro items; ± 0.02 is the indistinguishable-from-zero band.

Student	Factorization / objective	among- K	shuffled	Δ_q^{sh}
z1	single decision vector, bilinear scorer	.171	.147	+.023
zr	8 probes, token-level max-similarity	.157	.182	−.026
zr_set	+ $O(K)$ cross-attention over candidate tokens	.147	.151	−.003
ms_ctrl	single-set KD, steps-matched control	.124	.147	−.023
ms_zr_set	same- Z multi-set, explicit $\Delta\log\text{-odds}$ targets	.130	.130	+.000
zr (full depth)	8 probes, tapped at layer 28/28	.147	.156	−.008
Teacher	listwise decoder, options in context	.310	.208	+.102

.905/.873/.541/.826 at tap 20; Table 11). No further candidate-blind variants were run after this closure.

What *does* transfer into a candidate-blind state is worth stating precisely, because it is why accuracy hides the failure: the students inherit the teacher’s candidate priors and its calibration. Expected calibration error on the knowledge probe improves from .39 for an earlier baseline to .04, and the students are exactly set-stable where the teacher is set-fragile. A reader looking only at accuracy and ECE would conclude the factorization works.

Scope. This is a scoped empirical result, not an impossibility theorem. It covers the factorizations, objectives, depths, teacher, and budget listed above, at one backbone family and 1.7B with 12k steps. Two alternative explanations we cannot exclude: the set-conditioning path’s zero-initialized cross-attention stayed inert over 4k steps, so an optimization failure is a live alternative to an information-theoretic one; and the teacher’s own question-dependent component is small (.102), so the measurement has roughly nine points of dynamic range. What the result does establish is a design

Table 5: Letter-logit (N1) vs. contextual-candidate (N3) readout, two seeds each (identical config, seed=0 vs. seed=1), full-depth backbone (28 layers). Teacher = a fine-tuned decoder reading options in context. Δ_q^{sh} is the shuffled-question variant of Definition 4.1.

	N1 seed 0	N1 seed 1	N3 seed 0	N3 seed 1	Teacher
Δ_q^{sh} (primary)	.114	.104	.117	.094	.102
MMLU-Pro among- K	.325	.318	.314	.301	.310
SNLI	.838	.831	.863	.851	–
MNLI	.684	.664	.752	.719	–
ANLI	.377	.393	.431	.437	–
BoolQ	.769	.764	.737	.746	–
Reorder max Δp	.165	.165	.101	.109	0
Pairwise $\Delta \log\text{-odds}$	.463	.443	.126	.094	0

boundary we then acted on – and the diagnostic, Δ_q , transfers to any candidate-scoring architecture whether or not our particular negative does.

5.2 Contextual-candidate readout recovers the signal

We compare three non-generative readouts of the same decision pass, all with candidates rendered in the suffix: the **letter-logit readout** (N1: next-token probability over the rendered option letters – the ordinary multiple-choice interface, read directly rather than generated); the **candidate-blind semantic readout** (N2: a bilinear score between the terminal state h_D and an independently computed semantic embedding of the option text); and the **contextual-candidate readout** (N3, Section 3.2: a bilinear score between h_D and each option’s own contextual hidden state c_j^{ctx}).

The ordering is the interface-level result. Replacing the option’s contextual state with an independent semantic embedding of the same text costs 13 points of knowledge-probe accuracy (N2 .198 vs. N1 .325) and roughly half the question-dependent signal (Δ_q^{sh} .058 vs. .114). This is not the candidate-blind failure of Section 5.1 – N2 still runs the candidates through the language model and keeps a real Δ_q^{sh} – it is a narrower statement about the readout: an independently embedded candidate is measurably insufficient even when the candidates are in the computation. Table 5 then gives the matched two-seed comparison between the letter-logit and contextual-candidate readouts.

N1 and N3 land within seed noise of each other and of the teacher on Δ_q^{sh} (N1 mean .109, N3 mean .106, teacher .102): the contextual-candidate readout preserves essentially all of the letter interface’s question-conditioned signal while generating nothing. It does so with roughly a quarter of the letter interface’s order sensitivity (reorder Δp .10–.11 vs. .165; pairwise $\Delta \log\text{-odds}$ under candidate-set edits .09–.13 vs. .44–.46) and a consistent evidence-task advantage (SNLI +2, MNLI +6–7, ANLI +4–5 points, both seeds). Two honest caveats: the equivalence claim sits at the noise floor – the seed spread on Δ_q^{sh} is .01–.02 and the claimed gap is .003, so this is “indistinguishable at two seeds”, not a demonstrated equality – and order sensitivity is *reduced*, not eliminated; a permutation-invariant candidate-blind scorer gets exactly zero, so the honest position is that the contextual readout sits between the letter interface and a set-invariant scorer. Both arms train with per-step option shuffling, so part of the robustness is augmentation rather than architecture.

5.3 Why intermediate depth

Depth is the one lever that separates the two readouts, and it separates them in opposite directions. Reading the contextual-candidate readout at the tap-20 truncation rather than the full 28-layer backbone recovers evidence-task accuracy to within 1–3 points of a candidate-blind energy baseline on every evidence set (and +8 points on CLINC-150) while *keeping* the question-conditioned signal (Δ_q^{sh} .118 at tap 20 vs. .117 at full depth, 1.16 $\times$ the teacher’s). At the last layer the same readout loses 5–15 points of evidence accuracy relative to that baseline. For the candidate-blind state, the same depth change does nothing to Δ_q^{sh} at all (Section 5.1).

The conceptual point is more interesting than the flag: the final representation of a causal LM is optimized for next-token prediction, and it is not automatically the best representation for an evidence-grounded direct decision. Once the options are rendered inside the suffix, reading an intermediate layer is required to recover the backbone’s pretrained evidence reasoning. That mid-depth features

can beat last-layer features for classification is itself established [Gromov et al., 2024, Men et al., 2025, Tenney et al., 2019, Belrose et al., 2023]; what is specific here is that the choice is decisive for a deployed decision interface and interacts with the readout, not with the backbone. We chose one tap fraction (71%) and ported it across backbone sizes without a per-size sweep, which is a limitation.

5.4 Abstention: a rendered abstain option is a softmax-dilution bug

An early model rendered a literal “none of the above” line as one more option in the suffix and let it compete for softmax mass. This has a K -dependent pathology: the abstain option’s share dilutes as the candidate set grows, whether or not the true label is absent. Replacing it with the factored gate of Equation (1) fixes that essentially for free (Table 12): the K -sweep of null AUROC becomes monotone, MMLU-Pro false-abstention drops from .692 to .003 with accuracy rising from .133 to .353, and Banking77 goes from unusable (.063) to .579 – with Δ_q^{sh} unchanged (.118 vs. .122) and evidence tasks at the same level. The remaining costs are smaller and specific: intent/topic label spaces lose 2–7 points and order sensitivity rises slightly, because with no rendered null line the letter position is the only identity signal left in the suffix.

Three things that are easy to conflate should stay separate. The *rendered-null pathology* is a rendering artifact. The *exact pairwise non-null odds* property is an algebraic consequence of the factorization (Section 3.3). *Null calibration across K* is a measured property that the factorization makes achievable but does not guarantee – and it is partly a property of our training mixture, since under 1% of workflow rows naturally carry a null target, which is why null augmentation exists (Section 3.5). This is also the single-model, no-workflow-training configuration that scores .694 standard / .378 hard on JevBench (public subset, unranked) and is the base for every workflow-trained checkpoint in Section 6.

6 Training and scaling the decision interface

Having established what the representation must contain, this section asks how the interface behaves when it is trained and scaled: typed objectives (Section 6.1), the training distribution (Section 6.2), the size ladder (Section 6.3), probability quality (Section 6.4), and cost (Section 6.5). Two runs remained in flight at the time of writing (ts1b_semif, t11b_semif) and are omitted rather than reported provisionally; a Qwen3.5-9B checkpoint (t12) has hard-tier results only and appears solely in the resolution argument below, not in Table 6.

6.1 Typed objectives are contracts on the output distribution

The interesting result about Choice, Score, and Noul is not that they are three architectures – they are one backbone, one tap, and one readout – but that one readout supports genuinely different probabilistic semantics by changing only the target distribution and terminal head.

Matched fine-tunes (Appendix A.5 for full numbers) give two clean instances. Ordinal smoothing on Score changes *no* decision on JevBench’s 12 ordinal items relative to plain Choice while improving every probability-quality metric (held-out score NLL 2.07 $\rightarrow$ 1.23): a pure calibration change, which is exactly what an ordinal contract should buy and what a relabeling could not. The Bernoulli Noul head is exactly order-invariant by construction – there is no candidate order to be sensitive to – and is the first head in this project to clear an untouched external floor by a wide margin (PagerDuty .602 $\rightarrow$.886 against a .792 majority-class floor).

6.2 The training distribution controls capability

Adding rubric-conditioned workflow data (W) and the uncertainty corpus (U) to the evidence/knowledge mixture without adjusting sampling ratios costs evidence accuracy (CLINC-150 –11, TREC-fine –10 points), driven almost entirely by increased false abstention rather than lost discrimination. A sweep of E from .35 to .50 (Table 13) shows raising E to .45–.50 restores evidence accuracy to within 1 point of an E -only base and intent/topic accuracy to within 3–4 points, with W metrics unchanged within 1–4 points. Mixing, not architecture, caused the regression – and that is the general point worth carrying forward: the *distribution of decision types* is a first-order lever on capability, comparable in size to the architectural choices of Section 5.

Table 6: Scaling ladder. **JevBench: public 231-id subset, unranked** (72 standard items over 36 paraphrase-clustered states; 111 hard items). 95% intervals are cluster bootstraps over the paraphrase group field, 20,000 resamples (Section 4.5); they are wide and no unpaired difference smaller than them is a ranking. MMLU- K = MMLU-Pro among- K ; Δ_q^{sh} against a fine-tuned 1.7B teacher’s own .102; latency at $K=2$, same pod and build. Frozen = zero training, letter logits, 3 exemplars. Trained checkpoints: ts1b/tm1b (released as typical-small/typical-medium), ladder_8b, t11b, tm2 (Qwen3.5-4B, not released).

Model	JevBench std [95% CI]	JevBench hard [95% CI]	MMLU- K	Δ_q^{sh}	NLL	ms
Qwen3-1.7B (trained)	.694 [.569, .819]	.432 [.342, .523]	.343	.127	1.01	45
frozen, 3-shot	.528 [.389, .667]	.369 [.279, .459]	—	—	—	—
Qwen3-4B (trained)	.806 [.694, .903]	.423 [.333, .514]	.458	.193	1.02	57
frozen, 3-shot	.778 [.667, .889]	.441 [.351, .532]	—	—	—	—
Qwen3-8B (trained)	.667 [.528, .792]	.396 [.306, .486]	.290	.093	1.67	70
frozen, 3-shot	.556 [.417, .694]	.369 [.279, .459]	—	—	—	—
Qwen3-14B (trained)	.931 [.861, .986]	.450 [.360, .541]	.486	.235	0.95	59
frozen, 3-shot	.819 [.708, .917]	.559 [.468, .649]	—	—	—	—
Qwen3.5-4B (trained)	.861 [.778, .944]	.495 [.405, .586]	.429	.194	0.97	—

Null augmentation ($W:0.20$ of workflow rows duplicated with their gold removed, target $\emptyset$) does at the data level what a post-hoc calibration offset provably cannot do at the output level (Section 6.4): it lowers false-abstain rates and raises MMLU-Pro accuracy to the E -only base’s level, with the highest Δ_q^{sh} of any mixture tested. The uncertainty corpus improves likelihood (ChaosNLI NLL 1.27 $\rightarrow$ 1.00) while top-1 accuracy is flat or slightly lower, consistent with soft targets spreading probability mass rather than sharpening it.

DecisionMix v2 moves held-out rule-grammar and rubric-style generalization sharply within the generator’s own world (Appendix A.6) while leaving level-7 composition at chance. We treat the in-grammar half as an evaluation-protocol result rather than a generalization claim, since the held-out families and the training generator share one rule engine (Table 2, rows 4–6); the out-of-grammar half is the subject of Section 7.1.

6.3 The scaling ladder

Table 6 runs the identical recipe at four Qwen3 backbone sizes and once on Qwen3.5-4B, alongside a frozen three-shot control at each size (the base model reading next-token letter logits over the rendered options, no training at all).

Three readings, stated at the resolution the intervals support.

Standard-tier performance generally improves with scale, with an anomalous 8B point. We deliberately do not call this monotonic and do not fit a scaling law through it: it is a scaling *ladder* of single runs. Qwen3-8B is an outlier in both frozen and trained form under the identical recipe, and we found no tap or learning-rate setting that recovers it within budget; it should be read as a checkpoint-specific anomaly, not as evidence about 8B-parameter backbones. Three separate things are also being varied and should not be collapsed: backbone scale, the trained decision recipe, and the Qwen3-vs-Qwen3.5 pretraining generation – the single Qwen3.5-4B checkpoint reaches hard-tier .495, above every Qwen3 checkpoint we trained at any size, at a fraction of the parameters.

Training helps the standard tier; each individual rung is inside its interval. The trained model beats its frozen control on standard accuracy at all four rungs (+16.6, +2.8, +11.1, +11.2 points), which is consistent in sign at every rung, but no single rung’s gap clears its own 95% interval, and a paired bootstrap at 14B gives $-.111 [-.250, +.014]$ for frozen-minus-trained ($p=.10$). The pattern is real; any individual number is not a measurement of its size.

Per-decision latency is roughly flat across the ladder (45, 57, 59 ms at $K=2$ for 1.7B/4B/14B), because a decision’s cost is dominated by the cached-state pass and a short suffix rather than by backbone depth (Section 6.5).

The hard tier cannot resolve neighbouring rungs at all. A Qwen3.5-9B checkpoint trained with the same recipe scores exactly 55/111 on the hard tier, the same count as the Qwen3.5-4B checkpoint

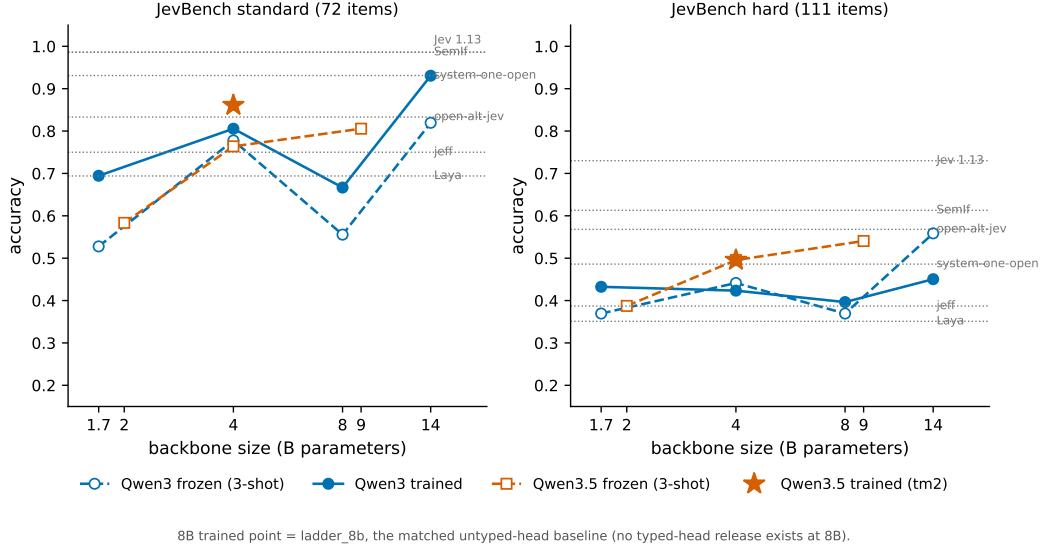


Figure 3: JevBench standard and hard accuracy vs. backbone size for Qwen3 (1.7/4/8/14B) and Qwen3.5 (4B; 2B/9B points from frozen controls where locally available); frozen-with-3-shots dashed, trained solid. **Public subset, unranked**; see Table 6 for the intervals, which are wide.

in Table 6. This is not a tie between equal models: the two disagree on 26 individual items (13 in each direction) and not one of their 111 probability vectors matches. McNemar’s exact test gives $p = 1.000$ with a paired 95% interval on the difference of $[-.090, +.090]$. The correct reading is that *this benchmark is uninformative about 4B-vs-9B at this sample size*, not that the two models perform equally, and the same caution applies to any adjacent pair of rungs in Table 6.

The hard tier also moves in the opposite direction from the standard tier, which is the subject of Section 7.

6.4 Probability quality is trainable, and task-conditional

We do not claim the model is “calibrated”. The result we can support is more specific and more useful: *probability quality is trainable, and what a good calibration looks like is conditional on the decision type and task family*.

A single global knob cannot serve two regimes. A single global temperature and null-logit offset, fit jointly on held-out soft-target rows, moves in *opposite* directions for an evidence/knowledge-only checkpoint and a workflow-trained one. For the former it reduces false abstention and repairs CLINC-150 (false-abstain .053 $\rightarrow$.022); for the latter the same fitting procedure pushes the offset to +3.0, which repairs typed-decisions NLL (1.95 $\rightarrow$ 1.19) but collapses intent/topic coverage to 5–9%. Calibration for this class of model has to be per decision-type or per family, which is what null augmentation (Section 6.2) and per-type target distributions (Section 6.1) supply at the data and objective level instead of at the output.

The selection criterion, not capacity, was the binding constraint. Under an accuracy-shaped selection criterion, held-out soft-target NLL got *worse* with backbone scale: 2.03 at 1.7B, 2.15 at 4B, 2.87 at 14B. Selecting checkpoints on held-out uncertainty-and-curriculum NLL instead, combined with a training-time Brier term rather than only post-hoc temperature fitting, takes the 14B model’s held-out score NLL from 2.87 – the worst point on the entire ladder – to 0.95 at the same backbone size and the same capacity. This is the largest single lever on soft-target probability quality found in this project, and it is an objective-and-selection lever, not an architectural or capacity one.

Where it does not hold. Hard-tier probability quality stays poor after every fix (Brier .85 $\rightarrow$.66 at 14B against a frozen control’s .60), and calibration on out-of-distribution label spaces is not established by any measurement here. Calibration in this paper is an evaluated axis with a task-and-type index, not a property of the system.

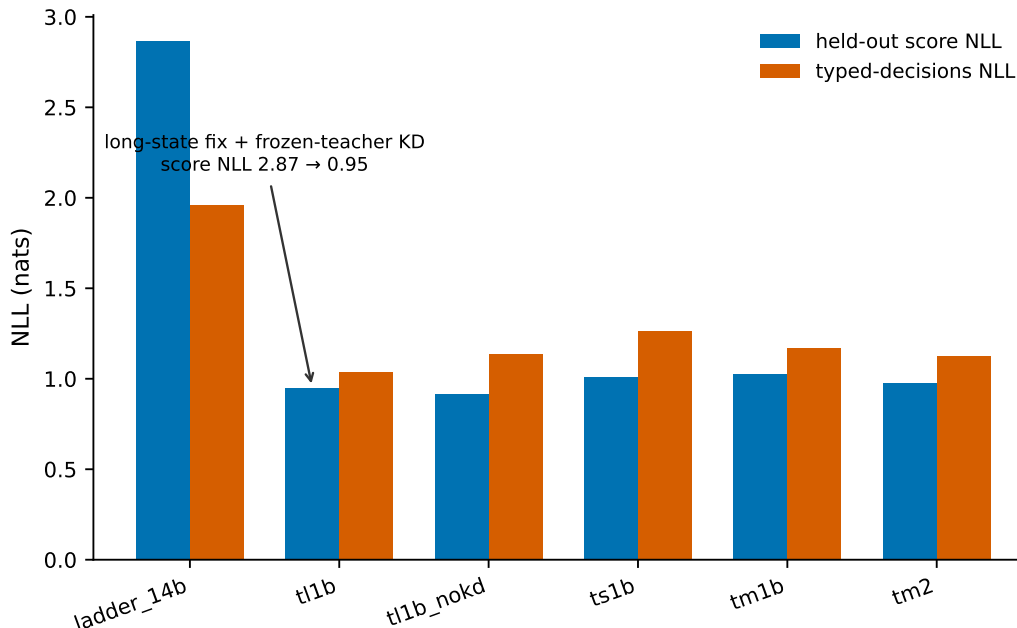


Figure 4: Held-out score NLL and typed-decisions NLL (held-out sets, not JevBench) across the calibration-fix sequence (ladder_14b $\rightarrow$ t11b/t11b_nokd/ts1b/tm1b/tm2), showing the 2.87 $\rightarrow$ 0.95 change in held-out soft-target likelihood at fixed backbone size.

6.5 Cost

We report this as an engineering measurement, not a benchmarked systems result: every number is a single-stream, Python-level research harness with no optimized-runtime or saturated-server baseline (Appendices A.2 and A.3 give the full K -sweep and the serving-path fix, which we keep deliberately separate from these numbers).

On one H100 (same pod and build, 256-token state), a single decision costs 45, 56, and 60 ms at 1.7B, 4B, and 14B at $K=2$ – an $8\times$ larger backbone is $1.3\times$ slower per decision, because the cached state and a short suffix dominate cost – rising to 106, 118, and 157 ms by $K=256$. The honest scientific claim is therefore: under this single-stream setup, direct readout avoids answer-token decoding and reuses the state prefix, and latency stays in the tens of milliseconds across the tested sizes, with model size and suffix length still materially affecting absolute cost. The structural claim for M queries against one cached state is $O(x) + M \cdot O(q + A)$: amortized in the state, *not* independent of candidate tokens. Native choice is the cheapest path for a single decision at $K \leq 32$; a candidate-blind energy-scoring front end that pre-filters larger sets to a bounded shortlist is cheaper in throughput mode at $K \gtrsim 32$ – the one place the rejected factorization of Section 5.1 remains useful, precisely because shortlisting does not require question-conditioned capability.

7 Where direct decision fine-tuning fails

The same recipe that improves standard bounded decisions does not solve hard compositional ones, and on that tier a frozen backbone with a few exemplars is not beaten by anything we trained. This section reports that boundary, the confounds attached to it, and one diagnosed data defect whose causal contribution we could not establish.

7.1 The hard compositional tier

Breaking JevBench’s hard tier down by sub-family (Figure 5, Appendix A.9) localizes where training helps and where it does not. Held-out rubric styles, families, and grammars *inside* our own generators’ distributions move sharply under workflow training (Section 6.2, Appendix A.6), but none of it

Table 7: 14B comparisons. **JevBench: public 231-id subset, unranked**; 95% cluster-bootstrap intervals over the paraphrase group field ($n_{\text{eff}}=36$ standard states, $n=111$ hard items). `t11b_nokd` is identical to `t11b` in every flag except `-distill_beta 0`. Paired per-item bootstraps for the three comparisons that matter are in the text; marginal intervals overlap heavily and should not be differenced by eye.

	ladder_14b	t11b	t11b_nokd	Frozen, 3-shot
JevBench std	.875 [.792, .944]	.931 [.861, .986]	.917 [.833, .986]	.819 [.708, .917]
JevBench hard	.468 [.378, .559]	.450 [.360, .541]	.477 [.387, .568]	.559 [.468, .649]
JevBench Brier std (hard)	.173 (.85)	.175 (.66)	.127 (–)	.30 (.60)
long_policy (hard subfamily)	.053	.158	.211	–
Training val NLL	–	.438	.410	–
Held-out score NLL	2.87	0.95	–	–
Level-7 composition	–	.620	–	–

transfers to *level-7* composition – decisions requiring temporal arithmetic, unit conversion, expected-value reasoning, or multi-attribute trade-offs – which sits at chance for every checkpoint trained on this project’s data, including the 14B model that leads the standard tier. It matches the JevBench hard tier’s own temporal/probability/trade-off families.

This is not obviously a capacity story: the frozen 14B backbone with three in-context exemplars does no better on these specific families either. The reading we favor is data coverage – our generators produce boolean rule composition up to six levels deep and never produce the compositional operations level 7 and JevBench’s long_policy/temporal/probability/tradeoff sub-families require, and no model here acquires composition from data that does not contain it. We cannot exclude the alternative that a single constant-depth pass is the binding constraint; Section 8 treats that as an open question rather than a conclusion.

7.2 A frozen backbone with exemplars is not beaten on the hard tier

Table 7 compares the previous-recipe 14B checkpoint, the bundle that fixed the long-state rendering, dropped over-length rows, added DecisionMix v2, the typed heads, and calibration-shaped checkpoint selection, the same bundle with frozen-teacher distillation removed, and the frozen backbone with three exemplars and no training at all.

The frozen backbone leads the hard tier, and this survives a paired test. A paired cluster bootstrap on the per-item difference over the same 111 hard items gives frozen-minus-trained $+0.108$ $[+.027, +.189]$, $p=.015$ against `t11b`. Against `t11b_nokd` the same test gives $+0.081$ $[-.009, +.171]$, $p=.082$ – consistent in direction but not separable at this sample size. So the result is real for the checkpoint we released and directionally consistent for its matched sibling, and it should be stated at that resolution rather than as a uniform “fine-tuning hurts”. On the standard tier the ordering reverses (frozen-minus-trained $-.111$ $[-.250, +.014]$, $p=.10$).

What this supports is that *our current fine-tuning recipe does not preserve or improve the backbone’s hard-tier behavior*. It does not by itself show that direct readout cannot solve hard tasks, that one forward pass is insufficient, or that autoregressive reasoning is required; the rendering confound of Section 7.3 weakens any such causal reading further, since we did not sweep rendering for the frozen hard-tier control.

Frozen-teacher distillation: no detectable effect either way. A KL term against the *same* backbone’s own frozen three-shot letter distribution is the one component of the 14B bundle with its own matched control. On that pair, `t11b_nokd` is nominally ahead on hard accuracy (.477 vs. .450), long_policy (.211 vs. .158), training validation NLL (.410 vs. .438) and standard-tier Brier (.127 vs. .175), while `t11b` is nominally ahead on standard accuracy (.931 vs. .917). But the paired bootstraps are $-.027$ $[-.090, +.027]$ ($p=.45$) on hard and $+0.014$ $[-.042, +.069]$ ($p=.81$) on standard. **No effect of same-backbone frozen-teacher distillation is detectable at this sample size, in either direction.** The direction is consistent across several metrics and it is inside the noise; we report it as an uninformative control, not as a negative finding, and it is not evidence about distillation in general.

The ladder_14b $\rightarrow$ t11b standard-tier gain is likewise not clean (.875 $\rightarrow$.931; paired $+0.056$ $[+.000, +.125]$, $p=.13$). The probability-quality change in the same bundle is much larger relative

Table 8: Rendering alone, on frozen weights. Identical Qwen3.5 base checkpoints, identical public JevBench items (**public 231-id subset, unranked**), letter-logit readout in both columns; only the state/option rendering differs. Δ is a paired cluster bootstrap over the per-item difference on the same items, 20,000 resamples.

Tier	Backbone	our rendering	replicated rendering	paired Δ [95% CI], p
standard	Qwen3.5-2B	.583	.597	+.014 [-.194, +.208], $p=.95$
	Qwen3.5-4B	.764	.847	+.083 [+ .014, +.167], $p=.048$
	Qwen3.5-9B	.806	.931	+.125 [+ .056, +.208], $p<.001$
hard	Qwen3.5-2B	.387	.441	+.054 [-.045, +.153], $p=.33$
	Qwen3.5-4B	.495	.468	-.027 [-.117, +.063], $p=.64$
	Qwen3.5-9B	.541	.595	+.054 [-.027, +.135], $p=.22$

to its own noise (held-out score NLL $2.87 \rightarrow 0.95$, hard-tier Brier $.85 \rightarrow .66$) and is the part of that result we stand behind. One consequence worth recording for practitioners: our own pre-registered release gate for the large model was a point threshold (“hard $\geq .559$ ”) on a quantity with a ± 9 -point interval. A gate of that form is not decidable at this sample size and should be restated on a lower confidence bound, or on a metric with more items.

7.3 Rendering is a first-class experimental factor

How the state and the candidate set are rendered into the prompt is not a presentation detail. We have two independent demonstrations of its size, from opposite directions.

(i) Rendering alone, frozen model, no training. Scoring the identical public JevBench items with the identical frozen Qwen3.5 checkpoints, changing only the rendering to a replication of another published system’s format, moves standard-tier accuracy by up to 12.5 points (Table 8). No weights, no exemplar count, and no item set differ between the two columns, and the effect survives a paired per-item bootstrap at two of the three sizes.

The largest of these swings, 12.5 points on a frozen 9B checkpoint, is of the same order as the entire published standard-tier spread between leaderboard entries, and larger than any difference between our own model variants in Table 6. It is concentrated on the standard tier; the hard tier does not move separably at any size.

(ii) Rendering mismatch moves a held-out family by 19 points. On a purpose-built held-out long-state evaluation set ($n=605$, no exact-state overlap with either training corpus, scored with a 4,096-token window so that nothing is truncated at evaluation time; state lengths $p50 = 1,965$ tokens, max 2,843), switching the evaluation render against a model trained on the opposite render costs 14.5 points at 1.7B for the facts-first-trained model and 19.0 for the facts-last-trained one (Table 10). The evidence is fully visible in every cell; only its position in the render changes. At 14B the same switch costs 7.6 and 6.8 points – the robustness asymmetry visible at 1.7B essentially disappears with capacity, which is worth stating as a caution in its own right: the 1.7B pair alone would have supported a “facts-first training is more robust” claim that the 14B pair does not.

(iii) A benchmark number that is mostly rendering. The clearest illustration is our own worst number. `ladder_14b` scores **.053** on JevBench’s `long_policy` subfamily, which earlier drafts of this work read as a capability collapse on long states. The same checkpoint scores **.919** on 605 held-out long states rendered the way it was trained (Table 10). Its `long_policy` number was therefore never a pure capability measurement: it compounds genuine truncation damage (Section 7.4) with a render mismatch against the benchmark’s fixed format, on $n=19$ items. We state this about our own number, not about anyone else’s.

The implication is methodological and it applies to us first: **a benchmark score is a joint measurement of model capability and interface/rendering compatibility**, so rendering must be controlled when comparing architectures. We do not use this to dismiss any other system’s results – we cannot, because we have not run their rendering on their weights under a controlled protocol. What we can say is that a comparison between two decision architectures that does not hold rendering fixed is measuring something other than the architectures, and that our own frozen-vs-trained comparison

Table 9: The underpowered first attempt: matched facts-first vs. facts-last training arms, 1.7B, single variable, scored on JevBench. **Public 231-id subset, unranked, 95% cluster-bootstrap intervals.** Nothing here resolves. Table 10 is the same question asked with enough items.

	Arm A (facts-first)	Arm B (facts-last)	Test			
long_policy	.316 (6/19)	.105 (2/19)	Fisher exact	$p=.232$;	bootstrap	Δ
JevBench hard	.378 [.288, .468]	.396 [.306, .486]	opposite direction	$[-.053, +.474]$		
JevBench std	.750 [.625, .861]	.736 [.597, .861]	indistinguishable			

Table 10: The 2×2 : training render $\times$ evaluation render, on 605 held-out long states with no truncation at evaluation (4,096-token window; state $p50 = 1,965$, max 2,843 tokens). Diagonal cells are matched; off-diagonal cells are render mismatch. The **matched-vs-matched** comparison isolates the truncation effect. Majority-class floors: policy_permit .612 ($K=2$, $n=330$), action_select .233.

Trained on	Scored on	overall	policy_permit	action_select
<i>1.7B, single-variable matched arms</i>				
facts-first	facts-first (<i>matched</i>)	.942	.933	.953
facts-first	facts-last	.797	.788	.807
facts-last	facts-first	.640	.615 (<i>at floor</i>)	.669
facts-last	facts-last (<i>matched</i>)	.830	.842	.815
<i>14B replication (t11b_nokd vs. ladder_14b)</i>				
facts-first	facts-first (<i>matched</i>)	.997	.997	.996
facts-first	facts-last	.921	.933	.905
facts-last	facts-first	.851	.839	.865
facts-last	facts-last (<i>matched</i>)	.919	.930	.905

(Section 7.2) inherits exactly this weakness. It is also why the leaderboard context for JevBench lives in Appendix A.11 rather than in the body.

7.4 Where the evidence sits: a diagnosed data defect with a demonstrated effect

The long-state generator originally rendered supporting facts *after* the request text. Two facts about this are exactly measured and not in doubt. First, the tokenizer’s `truncation_side` is `‘right’`, verified empirically on the released tokenizer rather than assumed from configuration: the window keeps the start of the render and drops the end. Second, at the released 1,024-token training window this removed the supporting facts from **98.8%** of long-policy rows (100% at the 256-token window used earlier in the project). Every model trained on that version was optimized to answer long policies from states that no longer contained the evidence for the answer. The state-length histogram is Figure 6.

The first ablation we ran could not resolve it, and the design was the reason. We trained two 1.7B arms identical in every flag except whether `data_wf_long` renders its facts at 0% or 97.6% of the state text (same rows, same labels, matched state length, $p50 = 8,702$ characters in both arms), at `-max_state 1024` with `-drop_truncated` off in both arms so that truncation bites exactly as it did at release, and scored them on JevBench hard. That comparison is a null result (Table 9): long_policy .316 (6/19) vs. .105 (2/19), Fisher exact $p=.232$, and the hard-tier aggregate moves the other way. It is also, in hindsight, the wrong experiment – $n=19$, and scored in a *single fixed benchmark render*, so it confounded the mechanism under test with render compatibility.

The 2×2 resolves it. We scored each arm on 605 held-out long states under *both* renders, with no truncation at evaluation in any cell. Comparing each model in its *own matched render* removes the render-compatibility confound entirely and isolates what training on truncated states cost. Table 10 gives all eight cells.

At 1.7B the matched-condition gap is $+0.112$, **95% CI** $[+.077, +.147]$, $p = 5 \times 10^{-10}$ (per family: policy_permit $+0.091$ $[+.043, +.139]$, action_select $+0.138$ $[+.086, +.190]$). At 14B it is $+0.078$, **95% CI** $[+.055, +.100]$, $p = 7 \times 10^{-12}$, the smaller gap only because the facts-first 14B is

at ceiling (.997). The effect replicates at two scales an order of magnitude apart, and the mechanism in Section 3.5 stands: training on states from which the evidence had been silently truncated cost real capability, and re-rendering facts-first recovered it.

Three scope notes, because this is a claim we previously could not make. (a) Only the 1.7B pair is a single-variable ablation; the 14B pair differs in the rest of the release bundle as well, so it corroborates the direction and magnitude at scale rather than isolating the variable a second time. (b) Render mismatch is a genuine, separately measurable effect of comparable size (Section 7.3) – it is what dragged the facts-last arm to its majority-class floor in the off-diagonal cell – but it does not explain the matched-condition gap away. (c) The evaluation set was built for this measurement and shares a generator with training, so the claim is about the training defect, not about long-context capability in general.

The methodological lesson is sharper than the engineering one, and it is the reason we report the failed ablation alongside the successful one. The same question, asked on 19 items in a single fixed benchmark render, returned $p=.232$ and an aggregate pointing the wrong way; asked on 605 items under a 2×2 that separates the two mechanisms, it returned $p < 10^{-9}$ at two scales. The metric was the problem, not the mechanism. Two further transferable points: a fixed-length window interacts with *where* the load-bearing content sits and not only with how long the render is, so a truncation-length ablation alone would never have surfaced this; and silent truncation produces confidently wrong behavior by removing the evidence while preserving the labels [Sambasivan et al., 2021]. Full distributions are in Appendix A.12.

8 Discussion

A. Candidate dependence is not merely a scoring inconvenience. The standard reading of the cross-encoder/dual-encoder trade-off is a cost-accuracy curve: putting the candidate in the encoder costs a forward pass and buys some accuracy. Section 5.1 says something sharper for question-conditioned closed-book decisions. It is not that a candidate-blind state scores a few points lower; it is that under every factorization, objective, and depth we tested, its above-chance accuracy carries *no* question-dependent component at all. The candidates appear to need to participate in the language model’s own computation for that capability to be expressible. Two practical consequences follow. The reusable-state formulation a systems engineer wants is available for *shortlisting* (Section 6.5) – which does not require question-conditioned capability – and not for the decision itself. And any evaluation of a candidate-independent decision architecture should report Δ_q , or an equivalent, because accuracy and calibration will both look healthy while the question is being ignored.

B. Direct decision interfaces trade generative flexibility for structural control. What the interface buys is concrete: output bounded to a runtime-supplied set, no answer parsing, typed probability semantics with an explicit abstention event, and state-prefix reuse across many decisions. What it costs is equally concrete and we found all of it empirically. Candidate tokens must fit in the suffix, so the per-query cost is not independent of K . Choice remains order-sensitive – reduced to about a quarter of the letter interface’s sensitivity, not eliminated (Section 5.2). Large candidate sets need a second stage. And hard serial computation is unresolved. This is a trade, not a strict improvement, and the paper should not be read as claiming otherwise.

C. Fine-tuning reshapes capability, not only interface. It would be convenient if training a model to emit direct decisions changed only the output format. Section 7.2 says it does not: the same recipe that improves the standard tier leaves the hard tier behind a same-size frozen backbone with three exemplars. The lesson generalizes past our architecture – teaching a model a target output distribution can improve behavior inside that distribution while degrading capability outside it, and aggregate in-domain metrics hide this by construction. The counterpart, which is the same coin, is Section 6.2: the distribution of decision types in training is a first-order lever on capability, comparable in magnitude to the architectural choices this paper is otherwise about.

D. Probability quality is task-conditional. There is no evidence in this paper for a single universal “confidence” calibration of a decision model. The same global temperature-and-offset fit wants opposite corrections on an evidence-trained and a workflow-trained checkpoint (Section 6.4); ordinal targets fix Score’s likelihood without changing a single decision; a Bernoulli head fixes an invariance

problem that no amount of recalibration would; and hard-tier calibration stays poor after every fix. Calibration is best treated as a measured axis indexed by decision type and task family, and pursued at the data and objective level rather than through one scalar knob at the output.

How much computation must happen between state and decision? The decision pass computes at fixed depth with no generated token, so there is no mechanism for a sequence of dependent intermediate computations. Log-precision transformers of fixed depth are equivalent in expressive power to constant-depth threshold circuits [Merrill and Sabharwal, 2023a], which provably cannot solve certain sequential composition problems at any parameter count, while conditioning on self-generated intermediate tokens provably escapes that class [Merrill and Sabharwal, 2023b, Feng et al., 2023]. The families where our hard tier stays at chance – temporal arithmetic, unit conversion, expected value, multi-attribute trade-offs – are exactly the shape this picture predicts a single constant-depth pass should struggle with. We are careful not to convert that into an explanation of our result: Section 7.1 favors the data-coverage reading, the two are not exclusive, and the rendering confound of Section 7.3 weakens any causal claim built on the frozen comparison. What the theory does suggest is where to look next. If the gap is computational rather than distributional, the fix is an explicit intermediate-computation mechanism – more passes, or a small amount of scratch work before a final typed readout – rather than more parameters or more of the same data. The question this paper leaves sharpened is not whether generation is necessary in general, but *how much computation must occur between state and decision, and when it can remain non-generative*.

9 Limitations

Scope. Text-only, bounded decisions over candidate sets that fit in a short suffix. No planning, search, tool execution, or vision. The typed primitives cover categorical, ordinal, and Bernoulli contracts and nothing else.

Generalization. “Runtime-defined labels” means the axes of Table 2, not arbitrary unseen domains or ontologies. Transfer is strong on unseen label wordings and unseen label spaces inside trained domains, mixed on external workflow sets (several sit at or below their own majority-class floor for every checkpoint, which measures the set as much as the model), and absent on unseen composition grammar. Our rubric-conditioned generalization results share one rule engine between the training generator and the held-out families, so they are an evaluation-protocol result rather than a claim about general compositional rule execution.

Computation. Hard compositional decisions are unresolved at every size tested. Large candidate sets require a shortlisting front end. Per-query cost scales with candidate tokens, so the cached-state saving is real but bounded.

Probability quality. Calibration does not hold uniformly: not on the hard tier, not out-of-distribution, and not across decision types under one global knob. We report it as an axis, never as a property of the system.

Interface artifacts. Choice retains order sensitivity. The exact-odds property of Equation (1) holds for pairwise non-null candidates under the factored softmax only – the null gate is set-dependent by design, so the model as a whole is not independent of irrelevant alternatives. Part of the measured order robustness comes from per-step option shuffling during training, not from architecture alone.

Statistical power. Outside the explicit two-seed comparisons of Section 5.2, every number in this paper is a single training run; noise-floor measurements elsewhere in this project put NLI/BoolQ variation at ± 0.7 points and held-out label-space variation at ± 3 –5 points between identically configured seeds. JevBench intervals are ± 6 –13 points (standard, $n_{\text{eff}}=36$) and ± 9 points (hard, $n=111$). Several differences we report, including the frozen-teacher distillation control and the JevBench-scored truncation ablation, are inside those intervals and are reported as such. The 1.7B/4B/8B/14B ladder is entirely within the Qwen3 family, with a single Qwen3.5-4B checkpoint as a second-family data point, not a second ladder; Qwen3-8B is an outlier in both frozen and trained form and we did not find a configuration that recovers it within budget. Two checkpoints were still training at the time of writing and are omitted rather than reported provisionally, and a Qwen3.5-9B

checkpoint has hard-tier results only. We also note the converse of low power: neighbouring rungs of the ladder are not separable on JevBench’s hard tier at $n=111$ (Section 6.3), so the ladder’s value is its trend, not any pairwise ordering within it. Where a question mattered enough, the fix was to build an evaluation set large enough to answer it rather than to argue about the benchmark (Section 7.4).

Evaluation. Every JevBench number is the public 231-item subset and is not a ranked leaderboard entry; the 146 judge-scored items required for ranking are not public, so we cannot rule out that our public-subset numbers differ systematically from a full-benchmark ranking. Rendering and protocol move scores by more than the differences between many entries (Section 7.3), which is why we treat cross-entry comparison as context (Appendix A.11) rather than evidence. Several evaluation-only external sets rely on rule-based or model-assisted gold labels that we did not independently re-verify. Latency numbers come from a single-stream Python harness with no optimized-runtime or saturated-server baseline.

Reproducibility and licensing. Inference code, the released checkpoints, and every run manifest (harness commit, checkpoint hash, dependency lock) are public. Training code and several generated corpora carry licence constraints from their upstream sources; Appendix A.8 records provenance and licence per corpus, and Appendix A.13 records what is and is not redistributable.

10 Conclusion

We asked whether a causal language model can be used as a direct typed decision engine without generating an answer token, and what has to remain inside its computation for that to work. The central result is that the most reusable formulation – a candidate-blind decision state, computed before the candidate set is known – fails to preserve question-conditioned closed-book capability under the factorizations, objectives, and depths we tested, and fails in a way that raw accuracy and calibration both conceal. Keeping the candidates inside the causal computation and reading their contextual states against a cached state prefix recovers that signal, at roughly a quarter of the letter interface’s order sensitivity, and supports typed categorical, ordinal, Bernoulli, and abstention outputs with no answer token emitted.

The result is not that generation is unnecessary in general. The same fine-tuning that improves standard bounded decisions does not solve hard compositional ones, and on that tier a same-size frozen backbone read with three exemplars is not beaten by anything we trained. Along the way we found that rendering – where the evidence sits, how the options are written – moves a frozen model’s benchmark score by more than the architectural differences it is usually used to adjudicate, which is a reason to control it before comparing decision architectures at all.

That leaves a sharper question than the one we started with: how much computation must occur between a state and a decision, and when can it remain non-generative?

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A Appendix

A.1 Additional ablation tables and figures

Tables 11 and 12 give the full numbers for the candidate-blind closure (Section 5.1) and the rendered-null-line ablation (Section 5.4) summarized in the main text. Figure 6 gives the long-state truncation histogram referenced from Section 6.2; the serving-path engineering section and Figure 8 referenced from Section 6.5 follow in Appendix A.3.

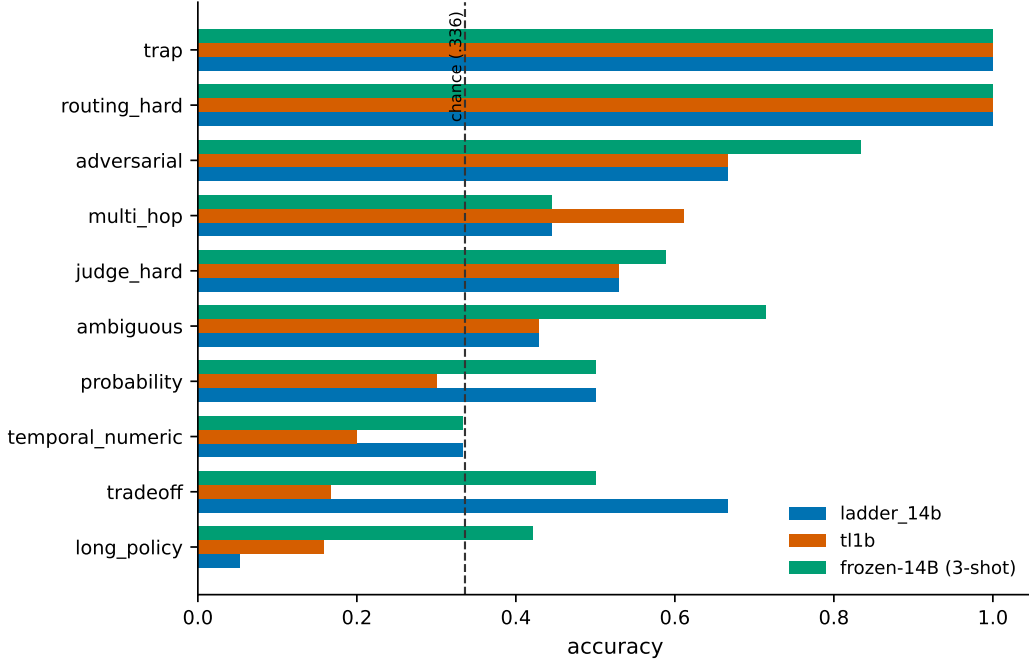


Figure 5: **Public subset, unranked.** Per-family JevBench-hard accuracy for ladder_14b, t11b, and the frozen-14B 3-shot control, sorted by t11b’s value, chance line at .336. Per-family values are re-derived directly from each run’s JSON artefact rather than transcribed from prose, and differ slightly from Section 7.2’s aggregate table in a small number of cells as a result; the JSON is the source of truth. Referenced from Section 7.1.

Table 11: Candidate-blind Z , closure at full depth vs. the original tap-20 student.

	Tap 20	Full depth (28)	Teacher
MMLU-Pro among- K / choices-only / shuffled-q	.157 / .166 / .182	.147 / .177 / .156	.310 / .222 / .208
Δ_q (choices-only) / Δ_q^{sh} (primary)	-.009 / -.026	-.030 / -.008	.088 / .102
SNLI / MNLI / ANLI / BoolQ	.905/.873/.541/.826	.881/.807/.462/.712	-

Table 12: Rendered $\emptyset$ line vs. the factored null, matched checkpoints (tap 20).

	Rendered $\emptyset$ line	Factored null (no rendered line)
MMLU-Pro false-abstain rate	.692	.003
MMLU-Pro accuracy	.133	.353
Banking77 (unseen labels) accuracy / false-abstain	.063 / .924	.579 / .135
CLINC-OOS accuracy / null AUROC	.639 / .840	.798 / .973
Δ_q^{sh} (unchanged)	.118	.122

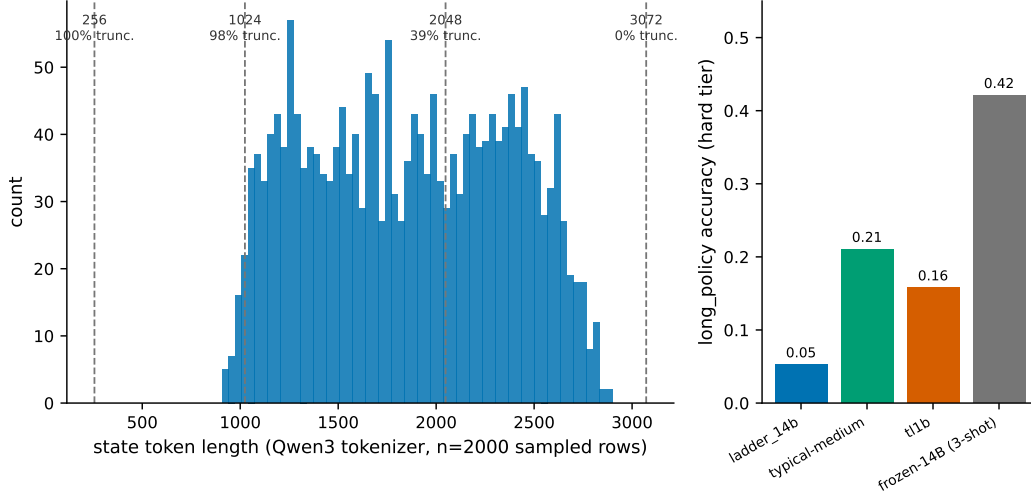


Figure 6: Left: histogram of `data_wf_long` state token lengths with truncation fractions at 256/1024/2048/3072 tokens. Right: `long_policy` (hard-tier) accuracy across the long-state bug/fix sequence.

A.2 Native choice vs. energy scoring vs. batched log-probability, full K -sweep

At $M=256$ queries against one cached 256-token state (transcribed from the run’s log; the corresponding `bench.json` was lost to an out-of-memory crash, so we report these as a table rather than a figure), native choice’s per-query marginal cost is 1.07, 1.55, 3.38, and 43.8 ms at $K=2, 10, 32, 256$; a candidate-blind energy-scoring path is flat at ≈ 1 ms across the same range; and a baseline that re-runs the backbone once per candidate costs 2.1, 4.6, 12.2, and 101.7 ms. Batched marginal cost at $M=256$ scales more steeply with both size and K than single-decision cost: 2.7 ms at $1.7B/K=2$ rising to 110 ms at $14B/K=256$, so the energy-scoring front end matters more at scale, not less.

A.3 Serving-path engineering

The public inference package initially deep-copied the entire cached prefix’s key-value tensors on every decision. Replacing this with a zero-copy strided view plus cached attention masks and position tensors reduced warm per-decision latency from 21–27 ms to **15.5–17 ms** (typical-small, 1.7B) and **19–21 ms** (typical-medium, 4B) on one H100 (Figure 8). We also attempted `torch.compile/CUDA-graph capture` for a further speedup: a single fixed input shape reached 8 ms, but probabilities shifted by up to .1 across the shape buckets the serving path actually sees, and the mutable key-value cache produced stale-buffer crashes under the standard capture-and-replay pattern; we reverted this and report it as a rejected optimization. Manual per-bucket graph capture remains an open path toward ~ 10 ms decisions but was not pursued further within budget.

A.4 Mixture sweep, full table

Table 13: Mixture sweep at 1.7B, matched steps, referenced from Section 6.2. “6A” is the $E:0.35/K:0.25/W:0.40$ point; “e45+null” adds `null_aug W:0.20` to $E:0.45/K:0.20/W:0.35$.

	Base (E only)	6A	e45	e45+null	long_{e45} (1024 tok)
CLINC-150 acc / false-abstain	.845/.05	.734/.17	.779/.10	.805/.08	.741/.13
MMLU-Pro among- K / Δ_q^{sh}	.353/.122	.330/.119	.338/.115	.353/.143	.329/.116
Held-out score NLL	—	2.03	1.52	2.04	1.26
JevBench std / hard	.694/.378	.750/.387	.833/.396	.806/.396	.806/.450

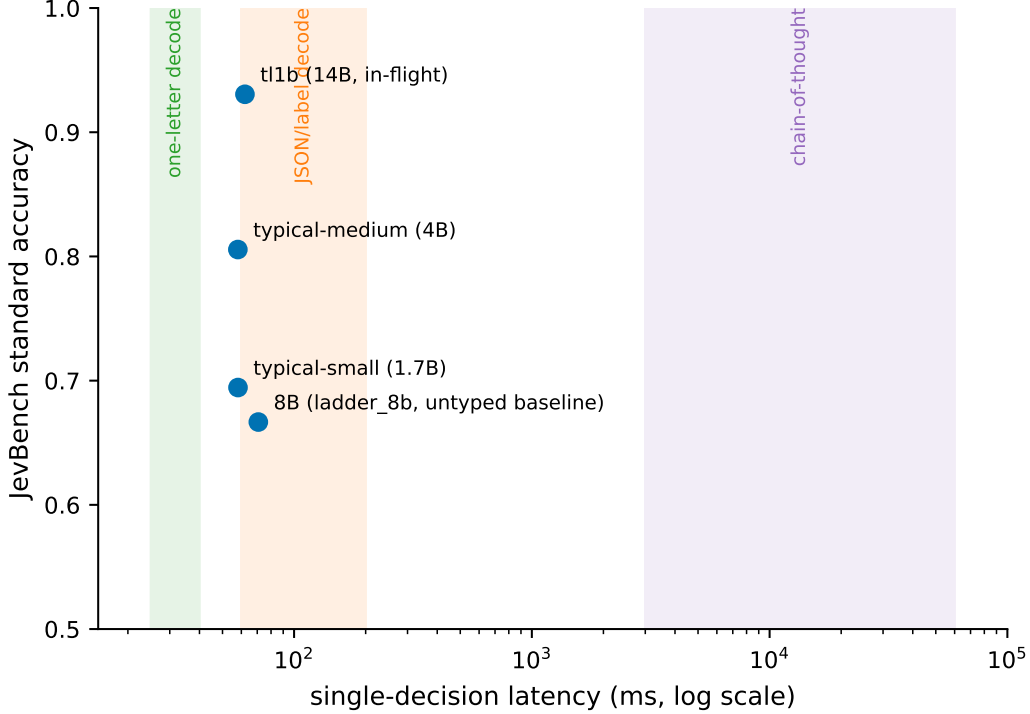


Figure 7: **Public subset, unranked.** JevBench standard accuracy vs. measured single-decision latency (ms, log scale) for our trained models, with generative-LLM decode-latency comparison bands. Referenced from Section 6.5.

Before the facts-first fix (Section 6.2), the long-state row (`longe45`, 1,024 tokens, 25% long-policy rows) alone raised `long_policy` from .16 to .37 and `multi_hop` from .17 to .28 relative to 6A, and gave the best soft-target NLLs of the sweep.

A.5 Typed primitives: parameterization, routing, and full results

Choice is the K -way readout of Section 3.2, unchanged: a hard one-of- K target trained with soft cross-entropy against Equation (1). **Score** is ordinal (severity, quality, urgency): plain Choice penalizes a one-level miss as much as an opposite-end miss, so we use an ordinal-smoothed target,

$$\tilde{t}_j \propto \exp\left(-\frac{|j-y|}{\tau}\right), \quad \tau = 0.7, \quad (2)$$

normalized over valid levels, $\tau=0$ recovering one-hot Choice – a pure target change, no architecture change. **Noul** is a yes/no proposition; rendered as 2-way Choice it is not guaranteed order-invariant, since “yes”/“no” still occupy two different suffix positions with two different contextual states. We give Noul a dedicated Bernoulli head reading the query alone, no candidate text,

$$P(\text{yes}) = \sigma(w_n \cdot h_D), \quad (3)$$

exactly order-invariant by construction, verified empirically as an exact-zero reversed-label control; the remaining mass $1-P(\text{yes})$ spreads over the other candidate(s), and the factored-null gate still applies.

Per-row routing, not a global flag. A row is Noul-eligible iff its candidate set is exactly {yes, no}; every other row scores as ordinary Choice, bit-identical to a run with Noul disabled. This boundary matters operationally: an early release candidate applying the Bernoulli head at the *model* level rather than per row collapsed to chance, because Choice- and Score-typed rows in the same batch were then scored with no candidate text at all. Fixing the routing per row (verified by 140 unit tests) restored it – a typed-decision interface is a contract on the input, not a labeling convenience, and violating it does not degrade gracefully.

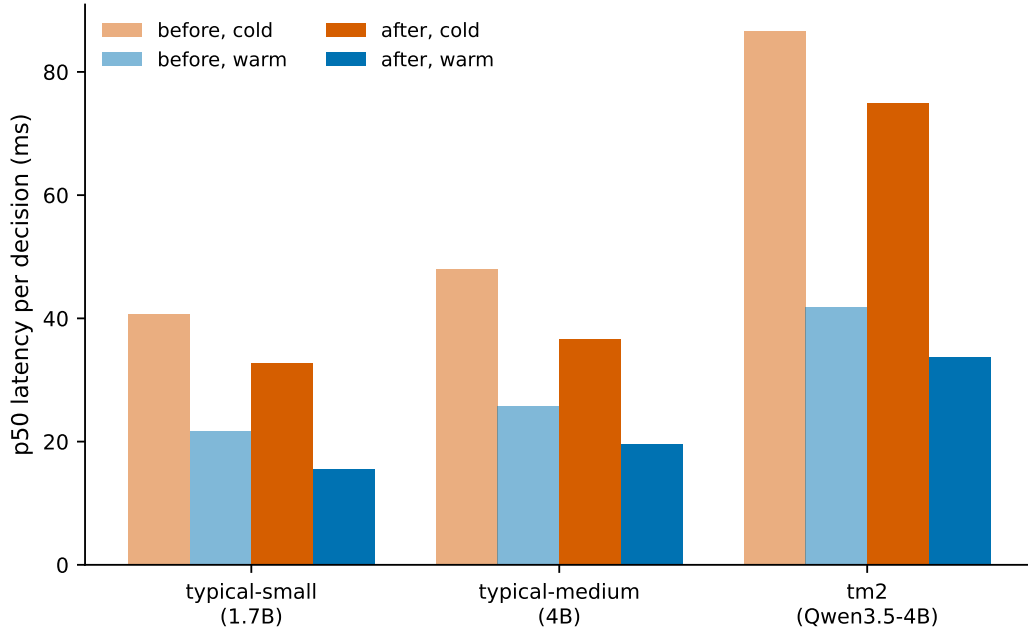


Figure 8: Per-decision p50 latency before/after the zero-copy prefix fix, cold vs. warm, for typical-small (1.7B), typical-medium (4B), and tm2 (Qwen3.5-4B), at $K=2$, 256-token state.

Table 14: Typed primitives vs. their K -way Choice control, matched fine-tunes, referenced from Section 6.1.

Score	K -way Choice	Ordinal-smoothed ($\tau=0.7$)
Held-out urgency: acc / NLL / ECE	.505 / 2.07 / .35	.508 / 1.23 / .19
JevBench hard: acc / Brier	.360 / .88	.459 / .70
JevBench ordinal items (12), decisions changed	–	0 of 12
Noul	2-way Choice	Bernoulli
PagerDuty (floor .792): acc / NLL	.602 / 1.04	.886 / .28
Reversed-label max $ \Delta P(\text{yes}) $	.55	0 (exact, all sets)

A.6 DecisionMix v2, full numbers

Held-out family/grammar/style accuracy moves .48/.51/.49 (matched control, never trained on data_wh) to .83/.88/.89 (DecisionMix v2), and rubric-flip accuracy moves .48 to .71, within the generator’s own rule engine (Section 7.2). Level-7 composition and typed-decisions NLL (2.06 $\rightarrow$ 1.71 with the uncertainty corpus) are reported in Section 7.2.

A.7 Full hyperparameters

Table 15 gives the released and candidate recipes’ exact configuration. All runs use AdamW, bf16, and LoRA rank 16 on the top eight kept layers of the truncated backbone; the family-balanced sampler and null augmentation are described in Section 3.5.

ts1b/tm1b additionally set null_aug W:0.20, ordinal_smooth 0.7 (Score), noul_head bern (routed per row, Section 3.3), null factored, and nc_render letters_nonull. tl1b adds -drop_truncated, frozen-teacher KD ($\alpha \cdot \text{CE} + \beta \cdot \text{KL}$, $T=2$, against the same backbone’s own frozen 3-shot letter distribution on 63k labelled rows), and checkpoint selection on held-out uncertainty-plus-curriculum validation NLL (-best_on) rather than in-distribution loss. Final fitted temperature and null offset for ts1b: $T=1.124$, offset 0 (post-hoc, applied at eval).

Table 15: Training configuration by checkpoint. Tap is given as layer / total layers of the base model. $E:K:W:U$ is the family-sampler weight per batch. Max state is the training-time token budget for the cached prefix.

Checkpoint	Backbone	Tap	Steps	Batch	$E:K:W:U$	Max state
typical-small-preview	Qwen3-1.7B-Base	20/28	12,000	64	.35:.25:.40:0	256
ts1b (typical-small)	Qwen3-1.7B-Base	20/28	12,000	64	.40:.15:.35:.10	1,024
tm1b (typical-medium)	Qwen3-4B-Base	26/36	12,000	64	.40:.15:.35:.10	1,024
ladder_8b	Qwen3-8B-Base	26/36	12,000	64	.35:.25:.40:0	256
ladder_14b	Qwen3-14B-Base	28/40	12,000	64	.35:.25:.40:0	256
t11b (candidate)	Qwen3-14B-Base	28/40	8,000	64	DecisionMix v2 + U , facts-first	3,072
tm2	Qwen3.5-4B-Base	$\approx 71\%$	12,000	64	.40:.15:.35:.10	2,048

A.8 Corpus construction detail

data_v5 (633,213 train / 4,001 val rows) covers evidence tasks and classification with a deliberately wide span of label vocabularies at low rows-per-source, so the model cannot memorize a fixed small set of label sets and must read the rendered options. data_kb (147,447 train / 3,001 val) is distilled from ARC-Easy/Challenge, OpenBookQA, CommonsenseQA, SciQ, QASC, LogiQA, AQuA-RAT, MedMCQA, and MMLU-auxiliary-train, with TruthfulQA-MC1 held out entirely and two items excluded from all MMLU-Pro numbers after a leakage audit against that evaluation set. data_wf (72,006 train / 3,000 val) is our own rubric-conditioned workflow generator, 15 families of Noul/Choice/Score decisions (policy checks, adequacy, support, fact verification, eligibility, routing, action selection, enumeration extraction, categorical judgment, tool selection, and four ordinal quality/severity/relevance/completeness scales, plus urgency); three whole families (eligibility, tool_select, urgency, one per type) are held out of training entirely, and 26,513 of the 72,006 rows belong to *rubric groups* – the same state and candidate set rendered under two or three different rubrics with different gold answers, built specifically to measure and train rubric dependence rather than state/candidate pattern-matching. data_wf_hf (86,600 train rows) adds three externally sourced sets in the same typed shape: cua-s1-forms (60,000 rows, procedural GUI-form option-scoring, MIT license, with a form-signature-disjoint 24,370-row test split and a 196-row real-usage demo split held out), systemone-lite-general (20,000 rows, rule-labeled typed decisions, MIT, with a 5,400-row hard split held out), and jev-4b-distill-data (6,600 rows, programmatic gold, Apache-2.0, with a 600-row adversarial split held out). data_wf_long (23,318 train rows) is a longer-state variant of the workflow generator for multi-paragraph policy documents, rendered facts-first (Section 3.5). data_wh (60,605 train / 3,000 val) is a second, independent rubric-conditioned generator (DecisionMix v2): a programmatic rule engine over 12 domains and six boolean conditions at seven levels of compositional depth, where levels 1–6 are trained and level 7 – compositions requiring temporal, unit-conversion, expected-value, or trade-off reasoning that the generator does not otherwise produce – is evaluation-only; every trained row belongs to a counterfactual rubric group. One whole domain/level pair, three whole domains, and two rubric styles are held out. data_u (31,029 train / 2,000 val) is a soft-target uncertainty corpus: the validation split of UNLI, the ambiguous rows of AmbiEnt (uniform target over the listed labels), and four synthetic generators with exact closed-form posteriors (partial-evidence combination, a noisy-sensor Bayesian update, ordinal confusion via an inverted confusion matrix, and conflicting-sources log-odds pooling), each with a held-out parameter range; ChaosNLI-ambiguity (MNLI portion) is evaluation-only.

A.9 Per-family JevBench-hard tables

Table 19 reports the hard-tier sub-families discussed in Section 7.1. As flagged in Figure 5’s caption, per-family values in the figure are re-derived directly from each run’s JSON evaluation artifact, which for ladder_14b’s temporal_numeric and probability sub-families differs from the corresponding prose in this project’s internal report (.333/.50 in the JSON vs. .20/.40 in the report’s own prose); we use the report’s own prose numbers below for t11b and the frozen control, since those cells do not exhibit the same discrepancy, and refer readers to the JSON-derived figure for ladder_14b’s two affected cells.

Table 16: data_wf: 15 families, 72,006 train rows, 3 held out (marked *). “In rubric groups” counts rows that are part of a same-state-same-candidates, different-rubric-different-gold group.

Family	Type	Rows	In rubric groups
policy_permit	Noul	9,000	5,205
adequacy	Noul	5,000	0
support	Noul	6,000	0
fact	Noul	5,000	0
eligibility*	Noul	1,500	851
routing	Choice	9,003	5,208
action_select	Choice	7,000	4,054
enum_extract	Choice	7,000	3,618
categorical	Choice	7,004	3,882
tool_select*	Choice	1,500	905
quality	Score	6,000	0
severity	Score	6,000	3,497
relevance	Score	3,998	0
completeness	Score	4,001	2,196
urgency*	Score	1,502	900

Table 17: data_wh (DecisionMix v2): 60,605 train rows across 12 domains $\times$ 6 boolean conditions, levels 1–6; level 7 (temporal/numeric/probability/trade-off composition) is eval-only.

Level	Train rows	Held-out grammar rows	Eval file	Rows
1	10,601	142	wh_heldout_family	422
2	10,600	141	wh_heldout_style	403
3	10,600	141	wh_heldout_grammar	816
4	10,601	140	wh_level7	880
5	10,602	142	wh_rubric_flip	2,206
6	10,601	141	wh_rubric_shuffled	2,437

A.10 Development chronology

Sections 5 to 7 are organized by claim. The experiments were run in a different order, and the order matters for reading the negative results, because several of them closed branches rather than improving a metric. The sequence was: (1) a candidate-blind decision-state branch (dual-encoder-style $Z(x, q)$ with an energy scorer over cached candidate embeddings), which produced the ladder of Table 4 and was closed by the shuffled-question control; (2) a redirect to a candidate-in-suffix decision pass, comparing letter-logit, candidate-blind semantic, and contextual-candidate readouts, which produced Table 5; (3) the two rendering/tap diagnoses – the last-layer tap and the rendered abstain line – which collapsed a planned two-regime (energy + native) architecture into a single model; (4) workflow and uncertainty corpora, the mixture sweep, and null augmentation; (5) the size ladder across Qwen3 1.7B/4B/8B/14B with frozen controls; (6) the 14B bundle (facts-first long-state rendering, -drop_truncated, DecisionMix v2, typed heads, calibration-shaped checkpoint selection, frozen-teacher KD) and its matched no-KD control; (7) the Qwen3.5 port; and (8) the post-hoc statistical work – cluster bootstraps, the matched truncation ablation, and the purpose-built held-out long-state set – which changed several claims in earlier drafts of this paper in both directions: it weakened the frozen-teacher distillation and standard-tier bundle results to the form they now take in Section 7.2, and it turned the long-state truncation diagnosis from an unsupported attribution into the measured 2×2 of Section 7.4.

A closure worth recording: the candidate-blind branch was not abandoned because a metric was disappointing. Each student closed 20–55% of the chance-to-teacher accuracy gap, which looked like progress. It was closed because the shuffled-question control showed that none of the closed gap was question-dependent. Had we selected on accuracy, we would have shipped that branch.

Table 18: data_u: 31,029 train rows, 100% soft-target.

Source	Train rows
Synthetic (4 closed-form generators)	28,560
UNLI (validation split only)	2,068
AmbiEnt (ambiguous rows)	401
By decision type: Noul / Score / Choice	23,403 / 7,225 / 401

Table 19: JevBench hard-tier sub-family accuracy, 14B. **Public subset, unranked**; per-family cells are 3–19 items each and carry no useful interval.

Sub-family	ladder_14b	t11b	Frozen 14B, 3-shot
long_policy	.05	.158	–
multi_hop	.44	.611	–
trap	1.00	1.00	–
adversarial	.67	.667	–
temporal	.20 [†]	.20	–
probability	.40 [†]	.30	–
tradeoff	.67	.17	–

[†] See discrepancy note above; the JSON-derived values for these two ladder_14b cells are .333 (temporal) and .50 (probability).

A.11 JevBench leaderboard context

This is context, not evidence, for the reasons set out in Section 4.5 and Section 7.3: protocols and renderings differ between entries, several entries have no method paper, the 146 private judge items are unavailable to us, the public standard tier has $n_{\text{eff}}=36$ independent states, and our own frozen-model rendering control moves scores by 12.5 points, which is larger than most of the spread in this table. We reproduce it so that readers can place our numbers, and we draw no capability ranking from it.

Table 20: Other entries scored on the same public 231 ids, from the harness’s own per-item files. **Public subset, unranked**, for all rows including ours. Majority/chance baselines: .311 standard, .284 easy, .336 hard. 95% cluster-bootstrap intervals for our own rows are in Tables 6 and 7; differences smaller than ± 6 –13 points (standard) or ± 9 points (hard) are not rankings.

Entry (backbone)	standard (72)	easy (48)	hard (111)
open-jev-deberta-v3-large (classifier)	.431	1.00	.378
GLiNER2	.639	1.00	.369
jeff (GLiFormer 400M)	.750	1.00	.387
Laya (ModernBERT)	.694	1.00	.351
open-alternative-jev (Qwen3.5-4B)	.833	1.00	.568
system-one-open (Gemma E2B LoRA)	.931	1.00	.486
SemIf (Qwen3.5-4B)	.986	1.00	.613
OpenJev (26B-A4B)	.972	1.00	.640
Jev 1.13.0 (closed)	.986	1.00	.730
Typical, Qwen3-14B (t11b)	.931	1.00	.450
Typical, Qwen3.5-4B (tm2)	.861	1.00	.495

A.12 Long-state truncation: distributions and the held-out evaluation set

Figure 6 gives the data_wf_long state-length histogram with truncation fractions at 256/1024/2048/3072 tokens, which is the exact measurement behind the 98.8% figure of Section 7.4, together with long_policy accuracy across the bug/fix sequence. The tokenizer’s truncation_side is ‘right’, verified empirically on the released tokenizer rather than assumed from configuration.

The held-out long-state evaluation set (`scripts/make_long_eval.py`, $n=605$) was built for the 2×2 of Table 10. It has zero exact-state overlap with either training corpus, with Case-stem overlaps dropped, and is scored at a 4,096-token window against states with $p50 = 1,965$ and $\max = 2,843$ tokens, so no item is truncated at evaluation in any of the eight cells. The two renderings differ only in whether the evidence block precedes or follows the request text; the items, labels, and candidate sets are identical. It shares a generator family with training data, which is why Section 7.4 scopes its conclusion to the training defect rather than to long-context capability in general. Significance for the matched-vs-matched comparison is a two-proportion test over the 605 paired items ($z = 6.2$, $p = 5 \times 10^{-10}$ at 1.7B; $p = 7 \times 10^{-12}$ at 14B).

A.13 Operations and reproducibility notes

Training and evaluation ran on rented RunPod H100 instances (SXM and NVL variants); a handful of infrastructure issues affected timing and are recorded here for reproducibility rather than folded into the results. A pod created with an SSH-only startup flag never reached a ready state on the account used and billed regardless (encountered repeatedly before being diagnosed); using a fixed exposed port at boot avoided it. Stopping a pod without an attached persistent volume wipes it, so checkpoints and manifests were always written to a network volume or pushed to the Hugging Face Hub before a pod was stopped. `tmux kill-session` does not terminate a `timeout`-wrapped child process, which once produced a stray duplicate training run sharing a GPU with the intended one; process matching over SSH must target the exact command, since a loose `kill` pattern can match and kill the SSH shell itself before the target process. Two pods sharing one Python environment on the same network volume corrupted it under concurrent dependency resolution; per-pod environments with dependency syncing disabled avoided this. Every run’s manifest records the harness commit, checkpoint hash, and dependency lock file used, and the JevBench per-item comparisons in this paper are all computed against the same fixed set of 231 public ids from one harness commit.

A.14 Full result tables

Tables 21 to 25 reproduce this project’s complete cross-checkpoint result tables at the time of writing, restricted to checkpoints that completed evaluation (in-flight checkpoints are omitted per Section 9).

Table 21: Evidence and knowledge-classification accuracy by checkpoint.

Model	Checkpoint	CLINC-150	TREC-fine	HWU64	20NG	SNLI	MNLI	BoolQ	ANLI
Qwen3-1.7B	ts1b	.801	.508	.761	.540	.894	.859	.824	.497
Qwen3-4B	tm1b	.847	.414	.769	.588	.909	.861	.843	.544
Qwen3-8B	ladder_8b	.769	.486	.740	.452	.853	.831	.822	.468
Qwen3-14B	t11b	.827	.508	.760	.690	.911	.868	.882	.583
Qwen3.5-4B	tm2	.795	.480	.717	.579	.900	.855	.840	.554

Table 22: Knowledge probe (MMLU-Pro among- K , Δ_q^{sh}) and TruthfulQA MC1.

Model	Checkpoint	MMLU-Pro among- K	Δ_q^{sh}	TruthfulQA MC1
Qwen3-1.7B	ts1b	.343	.127	.257
Qwen3-4B	tm1b	.458	.193	.408
Qwen3-8B	ladder_8b	.302	.093	.277
Qwen3-14B	t11b	.498	.235	.481
Qwen3.5-4B	tm2	.429	.194	.414

Table 23: Held-out workflow and external floors, full-row eval_wf pass.

Model	Held-out noul	Held-out score (NLL)	Held-out style	PagerDuty (.792)	jevlogs (.697)	Mind2Web (.427)
ts1b	.710	.522 (1.01)	.861	.560	.500	.351
tm1b	.811	.528 (1.03)	.861	.553	.480	.547
ladder_8b	.681	.543 (1.67)	.882	.321	.501	.404
t11b	.831	.621 (0.95)	.882	.805	.514	.564
tm2	.858	.586 (0.97)	.842	.757	.518	.497

Table 24: JevBench accuracy (Brier). **Public subset, unranked** (72 standard over 36 paraphrase-clustered states / 48 easy / 111 hard); see Tables 6 and 7 for intervals.

Model (checkpoint)	Standard (Brier)	Easy (Brier)	Hard (Brier)
Qwen3-1.7B (ts1b)	.694 (.40)	1.00 (.005)	.432 (.79)
Qwen3-4B (tm1b)	.806 (.30)	1.00 (.012)	.423 (.77)
Qwen3-8B (ladder_8b)	.667 (.55)	1.00 (.015)	.396 (.83)
Qwen3-14B (t11b)	.931 (.18)	1.00 (.005)	.450 (.66)
Qwen3.5-4B (tm2)	.861 (.22)	1.00	.495 (.72)
Qwen3.5-4B, frozen 3-shot	.764 (.36)	1.00	.495 (.60)
Qwen3.5-9B, frozen semif-render	.931 (.11)	1.00	.595 (.51)
Qwen3-14B, frozen 3-shot	.819 (.28)	1.00	.559 (.56)

Table 25: Single-decision latency (same-pod, same-torch-build; ms), $L_s=256$.

Model (checkpoint)	$K=2$	$K=32$	$K=256$	Peak memory $K=2 \rightarrow 256$ (GB)
Qwen3-1.7B (ts1b-shape)	45	46	106	6.4 $\rightarrow$ 9.3
Qwen3-4B (tm1b)	57	58	96	14.6 $\rightarrow$ 18.6
Qwen3-8B (ladder_8b)	70	71	126	29.3 $\rightarrow$ 34.1
Qwen3-14B (t11b)	59	62	156	51.6 $\rightarrow$ 57.5